



Information
Technology
Institute

ANALOG IC DESIGN

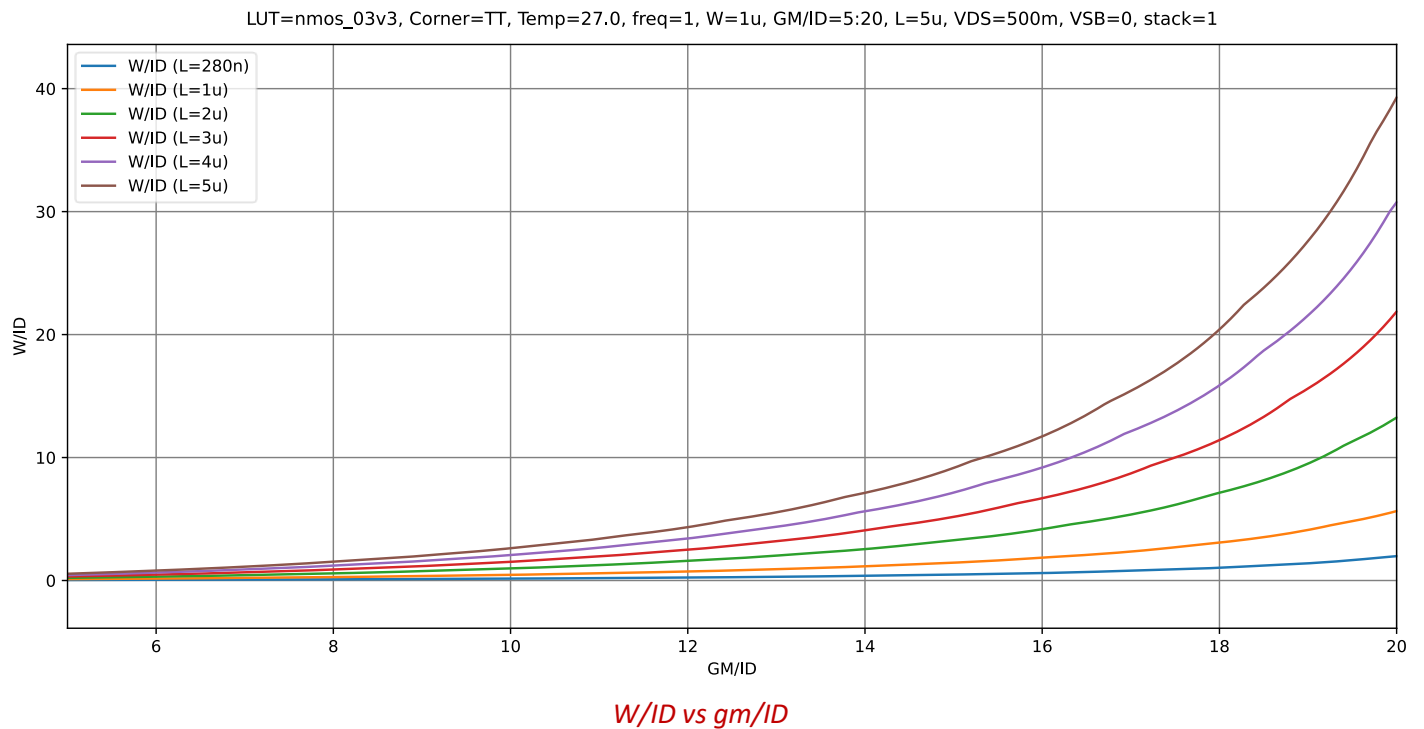
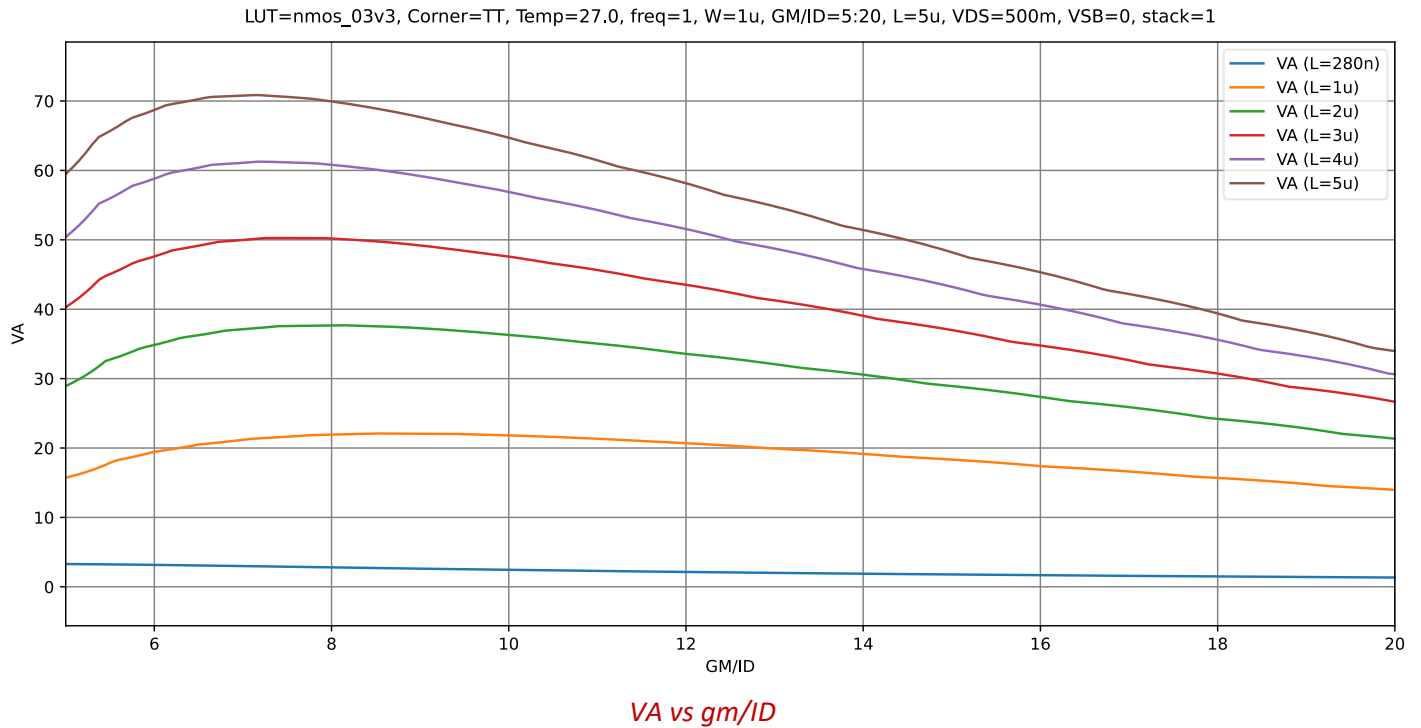
FULLY DIFFERENTIAL FOLDED CASCODE OTA

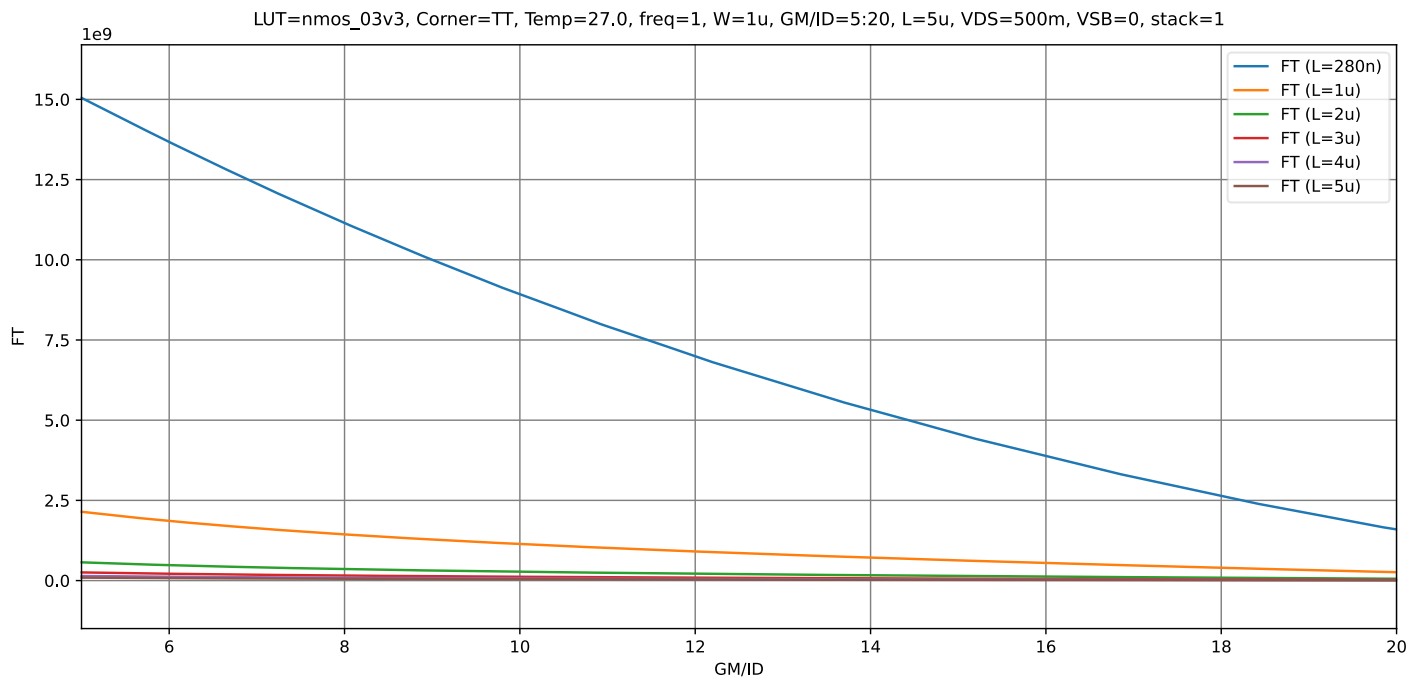
Submitted to: Dr. Hesham Omran

PART 1: gm/ID Design Charts

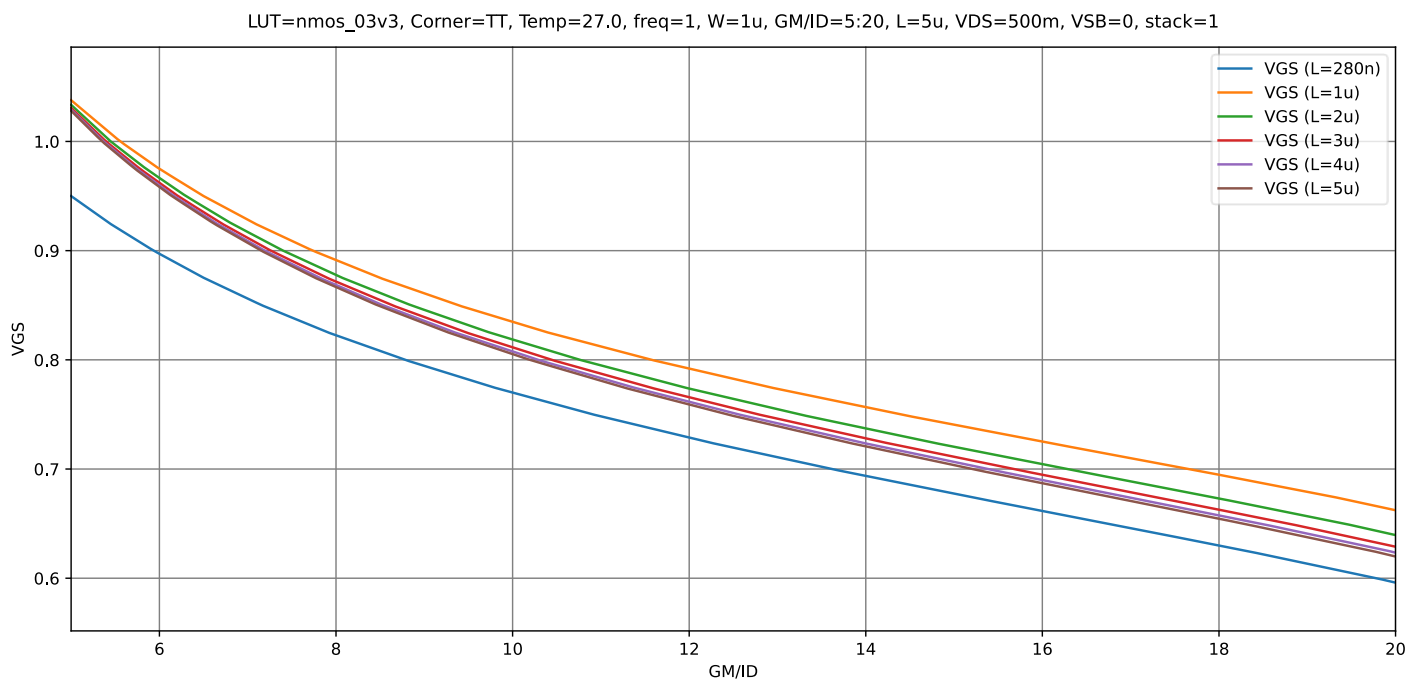
1) Using ADT Sizing Assistant, plot the following design charts vs gm/ID for both PMOS and NMOS. Set VDS to a reasonable value and L = min,1u:1u:5u.

NMOS:



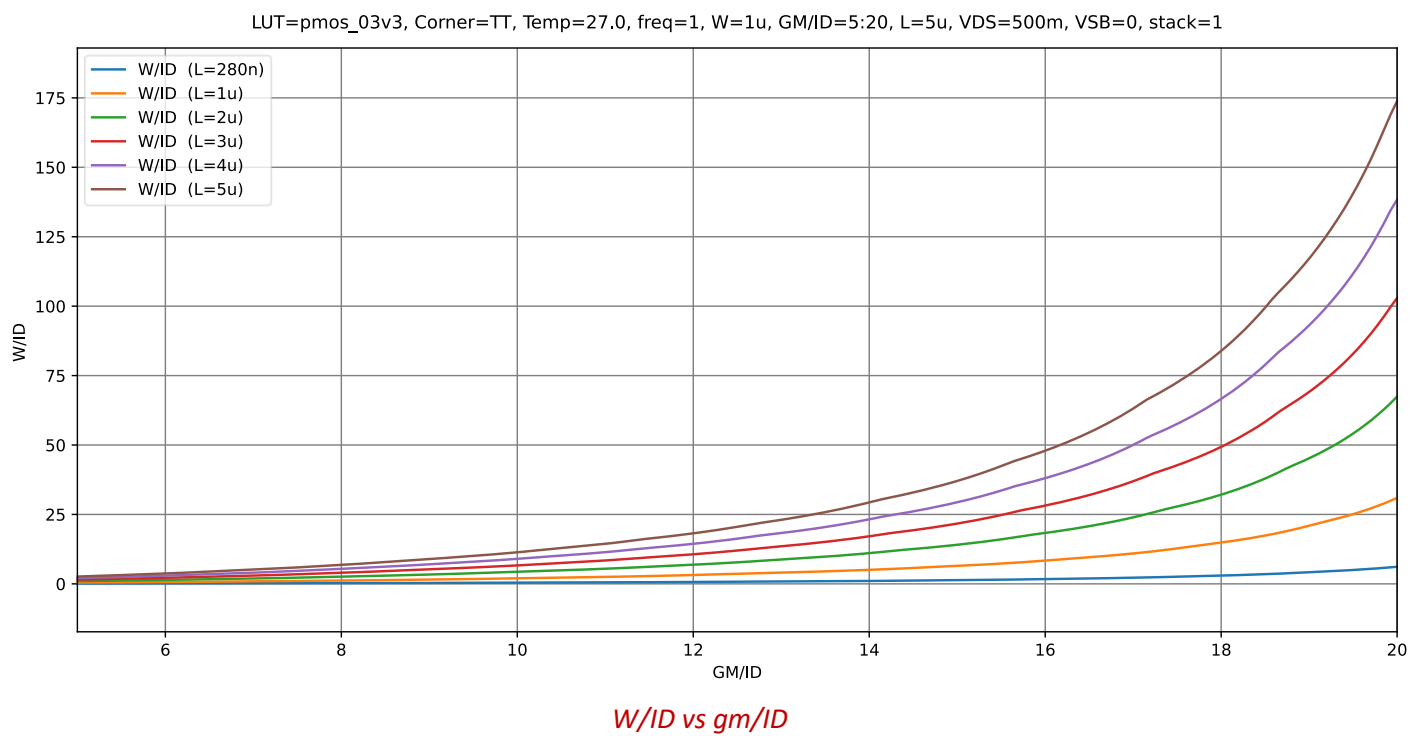
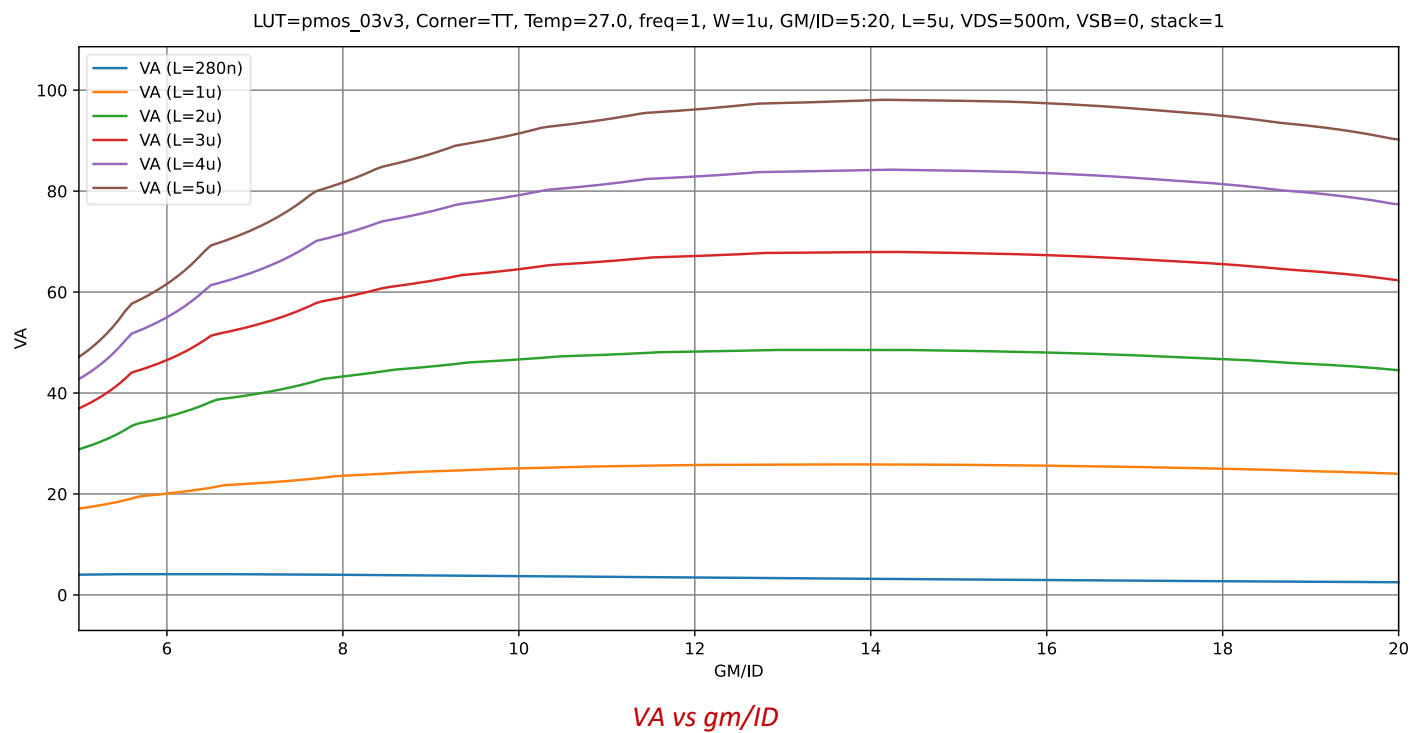


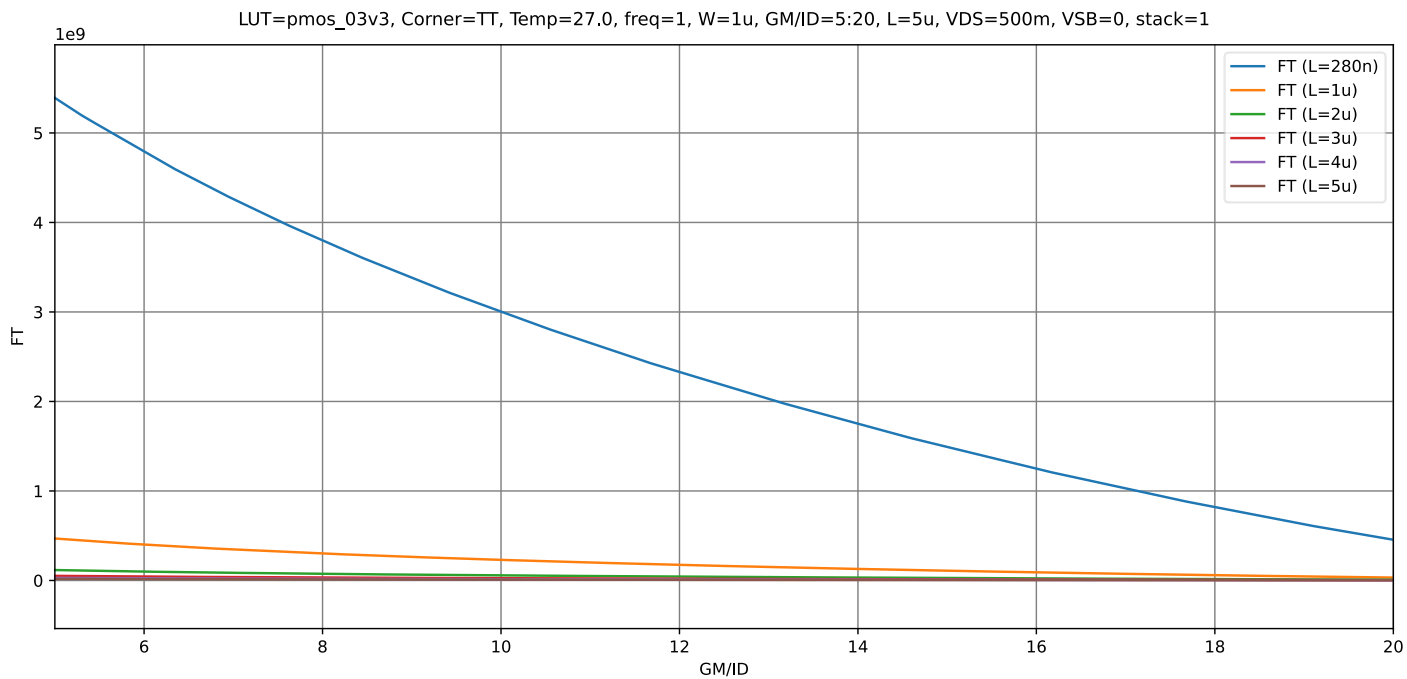
FT vs gm/ID



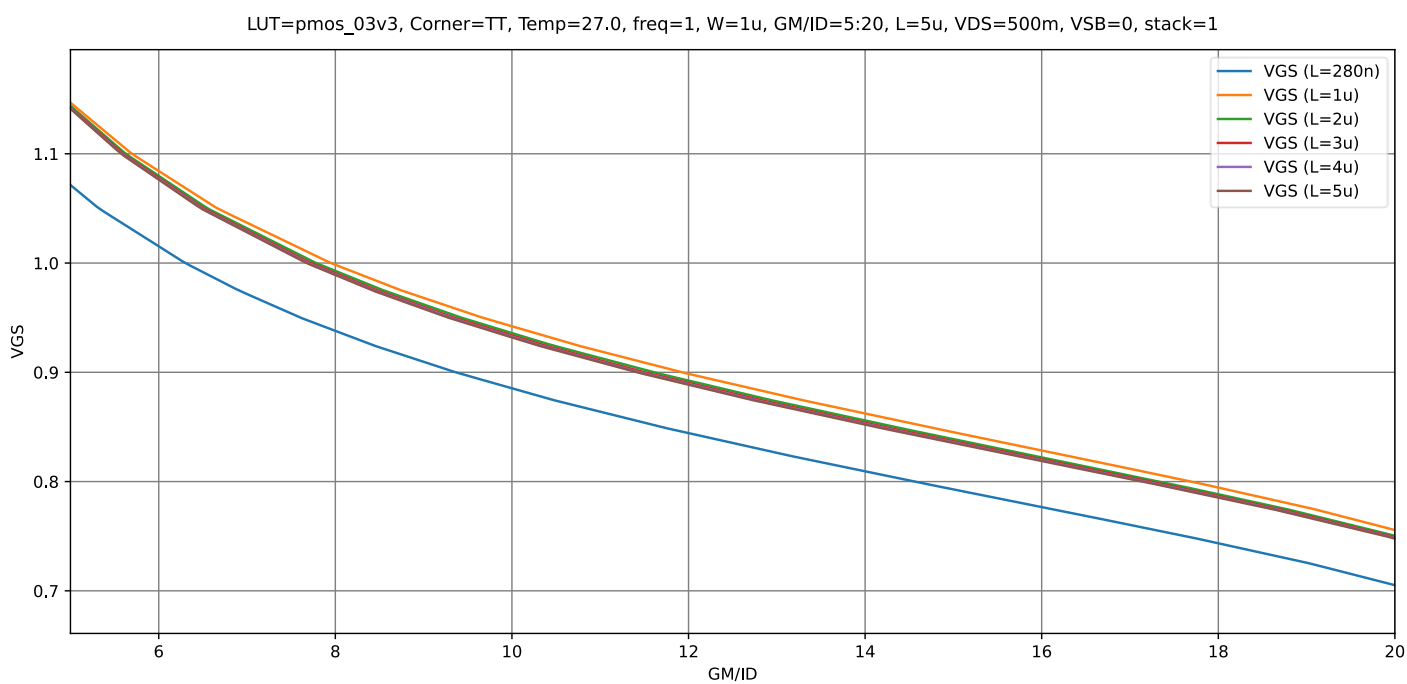
VGS vs gm/ID

PMOS:





FT vs gm/ID



VGS vs gm/ID

PART 2: OTA Design

Supply voltage	2.5V
Closed loop gain	2
Phase margin at the required ACL	$\geq 70^\circ$
CM input range – low	≤ 0
CM input range – high	$\geq 1V$
Differential output swing	1.2V _{pk-to-pk}
Load	500fF
DC Loop gain	60dB
CL settling time for 1% error	100ns

- 2) Convert the settling time spec to a spec on the closed-loop bandwidth (assume first-order system for simplicity).
- 3) From the closed loop bandwidth spec, calculate a reasonable bias current. The OTA current will be divided equally for the input pair (CS) and the cascode branches (CG).

$$4.605 * \tau = 100n$$

$$BW = \frac{1}{2 * \pi * \tau}$$

$$GBW = 3 * BW = 22MHz$$

$$gm_{input\ pair} = GBW * 2\pi * (C_L + C_F) = 207.3\mu S$$

AOL:

$$A_{OL} = \beta * A_{CL} = 3 * 1000 = 3000 = 69.54\ dB$$

1) Input pair sizing

- For the input pair use short L and bias it in MI or WI

Using minimum L and $gm/ID = 16$ then $ID = \frac{207}{16} = 13\mu A$

Let's keep a margin and choose current = **14 μ A**

After running the simulations W was increased to 32u to achieve gbw

LUT Settings

ID	14u	?
gm/ID	16	?
L	280n	?
VDS	0.7	?
VSB	0	?
Stack	1	?

Results:

	Name	TT-27.0
1	ID	14u
2	IG	N/A
3	L	280n
4	W	23.47u
5	VGS	774.3m
6	VDS	700m
7	VSB	0
8	gm/ID	15.96
9	Vstar	125.3m

2) Current sources

- For the current source transistors use relatively long L and bias them in SI

- PMOS Tail**

The tail current = $2 * 14 = 28\mu A$

Using $L=1\mu$ and $gm/ID = 10$

Note the pmos current sources in cascode branches will be matched to the tail but with half width

input pair

LUT Settings

ID	28u	?
gm/ID	10	?
L	1u	?
VDS	0.7	?
VSB	0	?
Stack	1	?

Results:

	Name	TT-27.0
1	ID	28u
2	IG	N/A
3	L	1u
4	W	54.63u
5	VGS	943.5m
6	VDS	700m
7	VSB	0
8	gm/ID	9.896
9	Vstar	202.1m

- NMOS source**

Using the same assumptions

Tail current source

LUT Settings

ID	28u	?
gm/ID	10	?
L	1u	?
VDS	0.7	?
VSB	0	?
Stack	1	?

Results:

	Name	TT-27.0
1	ID	28u
2	IG	N/A
3	L	1u
4	W	12.76u
5	VGS	836m
6	VDS	700m
7	VSB	0
8	gm/ID	9.877

NMOS current source

3) Cascode devices

- For the cascode transistors use moderate L and bias them in MI or WI

- PMOS**

Using $L=.5\mu$ and $gm/ID = 15$

ID	14u	?
gm/ID	15	?
L	.5u	?
VDS	0.6	?
VSB	.6	?
Stack	1	?
Results:		
	Name	TT-27.0
1	ID	14u
2	IG	N/A
3	L	500n
4	W	35.12u
5	VGS	1.087
6	VDS	600m
7	VSB	600m
8	gm/ID	14.1
9	Vstar	141.9m

PMOS cascode

- NMOS**

Using the same assumptions

After simulations to achieve Aol spec I used

$L=.51\mu$ and $W=10.71\mu$

ID	14u	?
gm/ID	15	?
L	.5u	?
VDS	0.6	?
VSB	.6	?
Stack	1	?
Results:		
	Name	TT-27.0
1	ID	14u
2	IG	N/A
3	L	500n
4	W	10.5u
5	VGS	937.9m
6	VDS	600m
7	VSB	600m
8	gm/ID	14.72
9	Vstar	135.9m

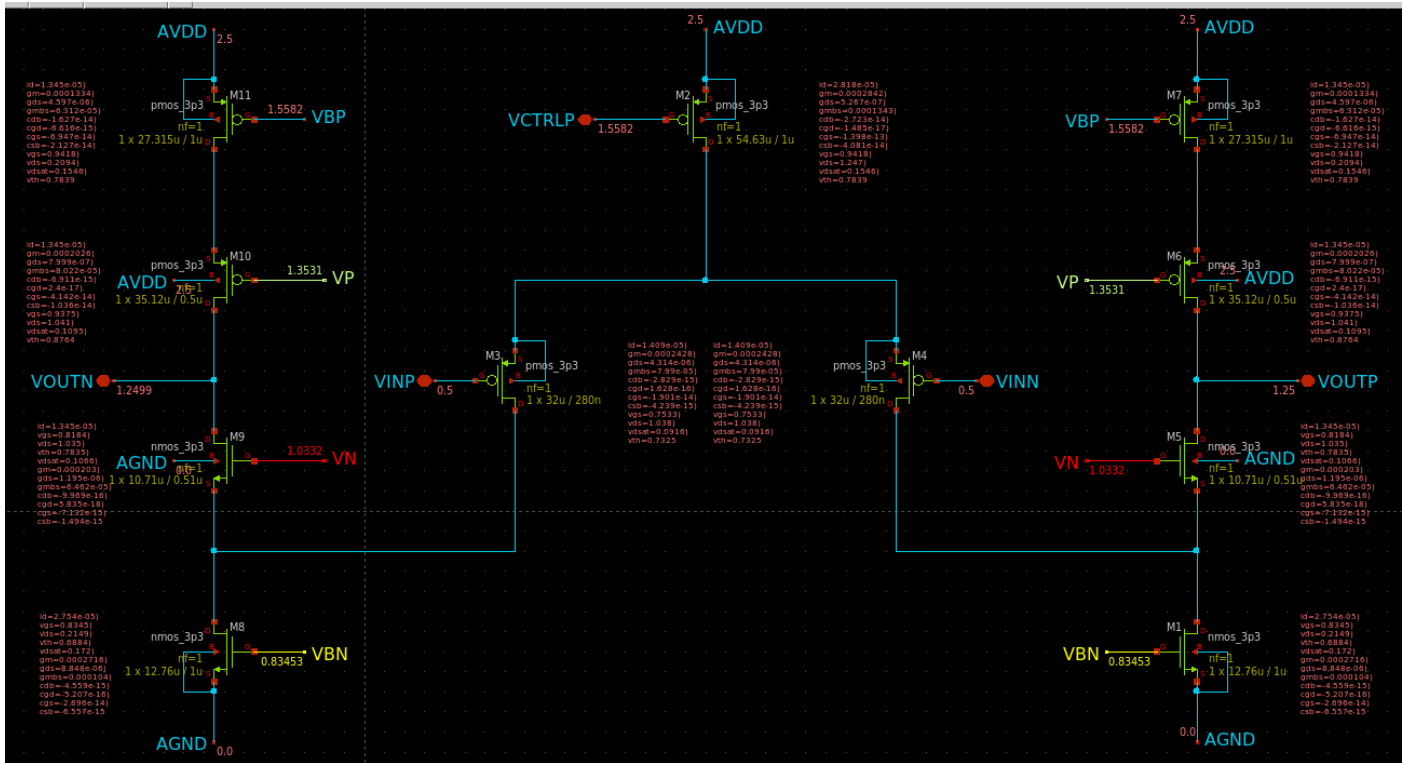
NMOS cascode

- 4) From the assumed V^* , select suitable biasing for the cascode transistors (VCASCP and VCASCN).

$$VCASCN(V_n) \approx V_{GSN} + V^* = .9379 + \frac{2}{10} = 1.138V$$

$$VCASCP(V_p) \approx V_{DD} - |V_{GSP}| - V^* = 2.5 - 1.087 - \frac{2}{10} = 1.213$$

Due to using of simple current mirrors so the current and the parameters will slightly differ so the biasing voltage will also differ



4) Cascode biasing circuits

After running OP and getting actual values of vgs then:

$$VCASCN (V_n) = 1.03V$$

$$VCASCP (V_p) = 1.35$$

- NMOS

We need vgs of this transistor to be 1.03V and to be matched to the cascode so it has the same L, I also used the same current for simplicity

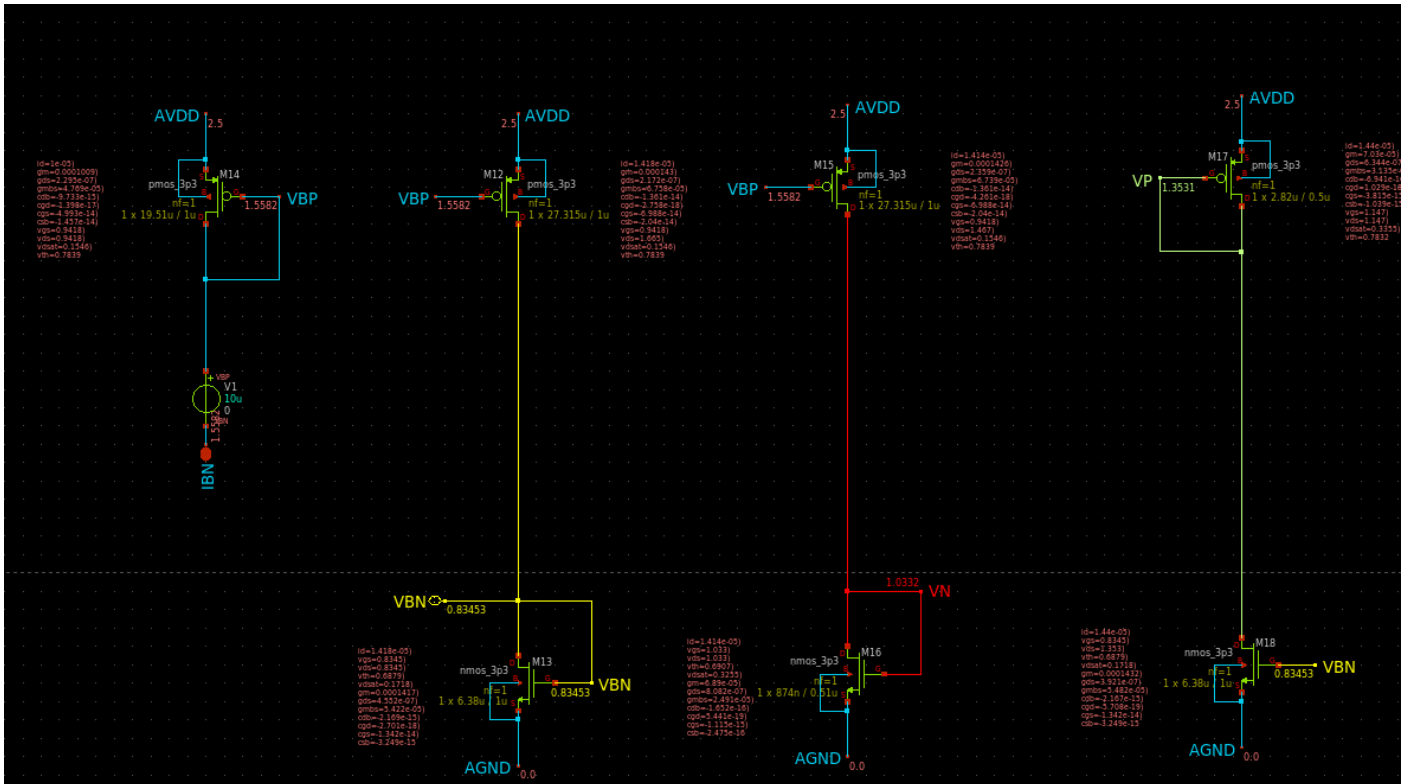
Also due to the simple current mirror there is an error, so doing a sweep on W around the ADT value the **Width = 874n**

LUT Settings		
ID	14.13u	?
VGS	1.03	?
L	.51u	?
VDS	vgs	?
VSB	0	?
Stack	1	?
Results:		
Name	TT-27.0	
1 ID	14.19u	
2 IG	N/A	
3 L	510n	
4 W	880n	
5 VGS	1.03	
6 VDS	1.03	
7 VSB	0	
8 gm/ID	5.087	
9 Vstar	393.1m	

- PMOS

We need v_g of this transistor to be 1.35V ($V_{GS} = 2.5 - 1.35 = 1.15V$) and to be matched to the cascode so it has the same L , I also used the same current for simplicity

LUT Settings		
ID	14u	?
VGS	1.14065	?
L	.5u	?
VDS	vgs	?
VSB	0	?
Stack	1	?
Results:		
	Name	TT-27.0
1	ID	14.01u
2	IG	N/A
3	L	500n
4	W	2.82u
5	VGS	1.141
6	VDS	1.141
7	VSB	0
8	gm/ID	4.976
9	Vstar	401.9m



	Input pair	Tail CS	NMOS CS	NMOS cascode	PMOS cascode	NMOS replica	PMOS replica
ID	14u	28u	28u	14u	14u	14u	14u
W	32u	54.63	12.76	10.71u	35.12u	874n	2.82
L	280n	1u	1u	.51u	.5u	.51u	.5u

PART 3: Open-Loop OTA Simulation (Behavioral CMFB)

- 1) We use ideal buffers to avoid loading the OTA output with the CM sensing resistors. Note that we don't need high gain in the CMFB loop (why?)

We don't need high gain in the CMFB loop because its role is only to regulate the output common-mode level within a tolerance, not to provide precision amplification — in fact, limiting its gain/bandwidth improves loop stability without degrading differential-mode accuracy.

1) OP

Vref = 1.25 (middle of the range)

```
s7
.control
save all

*=====
* This is a code that loops on all transistors in your schematic. If you
* have MOSFETs only in your schematic, set sym_loops variable to 1,
* comment the first sym vector, uncomment the second sym vector, and
* add the instance names of MOSFETs (ex: xm1 xm2 xm3 ....). If you want
* to use symbols, uncomment the first sym vector, comment the second
* sym vector, set loops variables to the number of symbols and MOSFETS
* , and add the instance names of symbols (ex: x1. x2. z1. z2. ...etc). This
* code relies on the order of elements inside the vectors rather than their
* names. There is echo line for testing the code.
* NOTE: To make a list using vectors, you must add a space after '(' and
* before ')' to avoid errors.
*=====

* == Vector For Using Symbols ==

set sym = ( x1. x3. )

* == Vector For Not Using Symbols ==

*set sym = ( )

* == Vector For MOSFETs ==

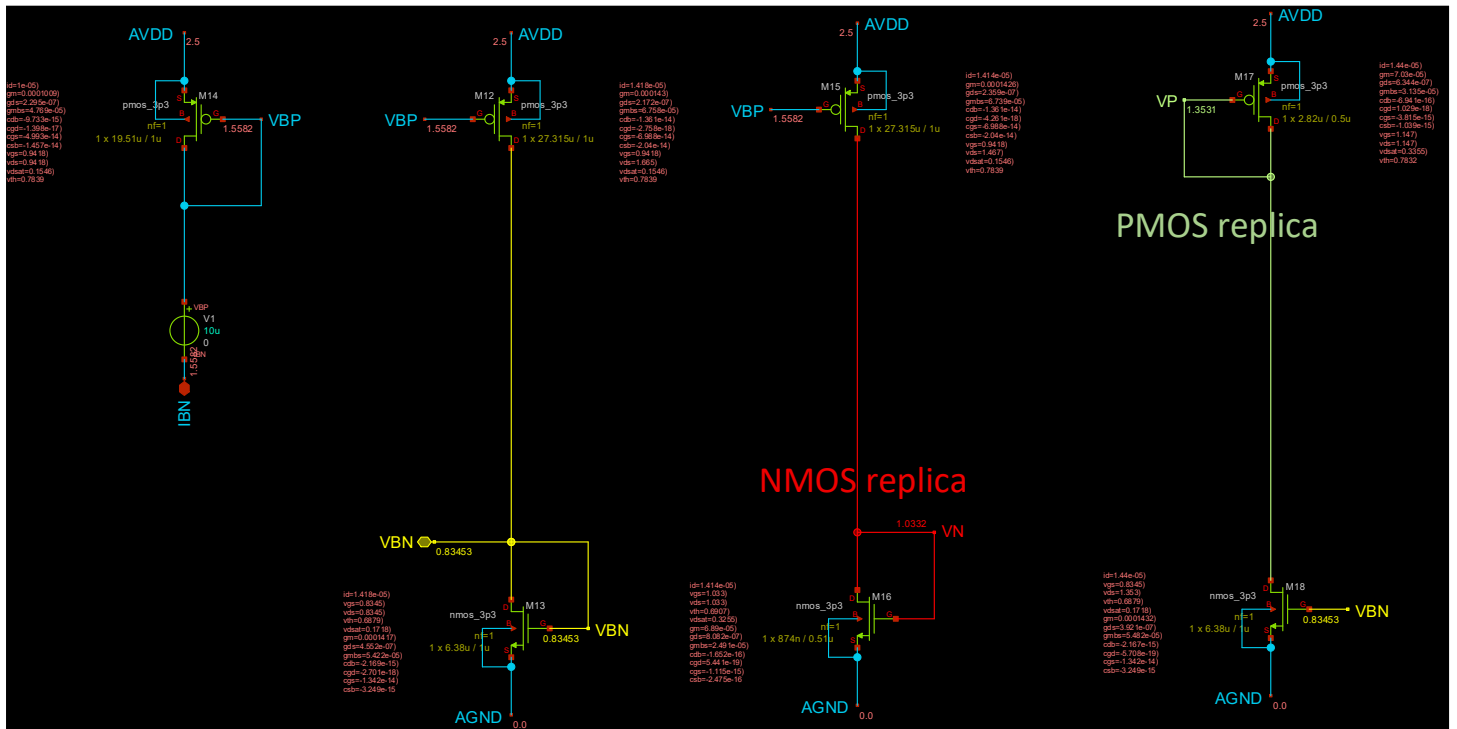
set mos = ( xm1 xm2 xm3 xm4 xm5 xm6 xm7 xm8 xm9 xm10 xm11 xm12 xm13 xm14 xm15 xm16 xm17 xm18 )

* == Initialize Counters For Loop ==

let i = 1
let x = 1
let sym_loops = 3
let mos_loops = 18

repeat $sym_loops
  dowhile x <= $mos_loops
    echo m.$sym[$i]$mos[$x].m0
    save @m.$sym[$i]$mos[$x].m0[id]
    save @m.$sym[$i]$mos[$x].m0[vgs]
    save @m.$sym[$i]$mos[$x].m0[vds]
    save @m.$sym[$i]$mos[$x].m0[vdsat]
    save @m.$sym[$i]$mos[$x].m0[vth]
    save @m.$sym[$i]$mos[$x].m0[gm]
    save @m.$sym[$i]$mos[$x].m0[gds]
    save @m.$sym[$i]$mos[$x].m0[gmb]
    save @m.$sym[$i]$mos[$x].m0[cgs]
    save @m.$sym[$i]$mos[$x].m0[cgd]
    save @m.$sym[$i]$mos[$x].m0[csb]
    save @m.$sym[$i]$mos[$x].m0[cdb]
    let x = x + 1
  end
  echo
  let i = i + 1
  let x = 1
end

op
write folded_ota.raw
.endc
```

region:

```
s3
control
  save all

  * --- Run operating point ---
  op

  * --- Constants ---
  let vtm = 0.026

  * --- Allocate vector for 12 MOS regions ---
  let region_vec = vector(18)*0

  * --- Loop over devices .xm1..xm12 ---
  let i = 1
  dowhile i <= 18
    let id = @m.xl.xm{$i}.m0[id]
    let vgs = @m.xl.xm{$i}.m0[vgs]
    let vds = @m.xl.xm{$i}.m0[vds]
    let vth = @m.xl.xm{$i}.m0[vth]
    let vdsat = @m.xl.xm{$i}.m0[vdsat]

    let subth_low = vth - 10*vtm

    let region = 0
    if (vgs >= vth & vds > vdsat)
      let region = 2 ; saturation
    else
      if (vgs >= subth_low & vgs < vth)
        let region = 3 ; subthreshold
      else
        if (vgs >= vth & vds < vdsat)
          let region = 1 ; triode
        else
          if (id < 6.3e-9 | vgs < 1e-10)
            let region = 0 ; cutoff
          else
            let region = 1
          end
        end
      end
    end

    * store result in vector
    let region_vec[{i-1}] = region

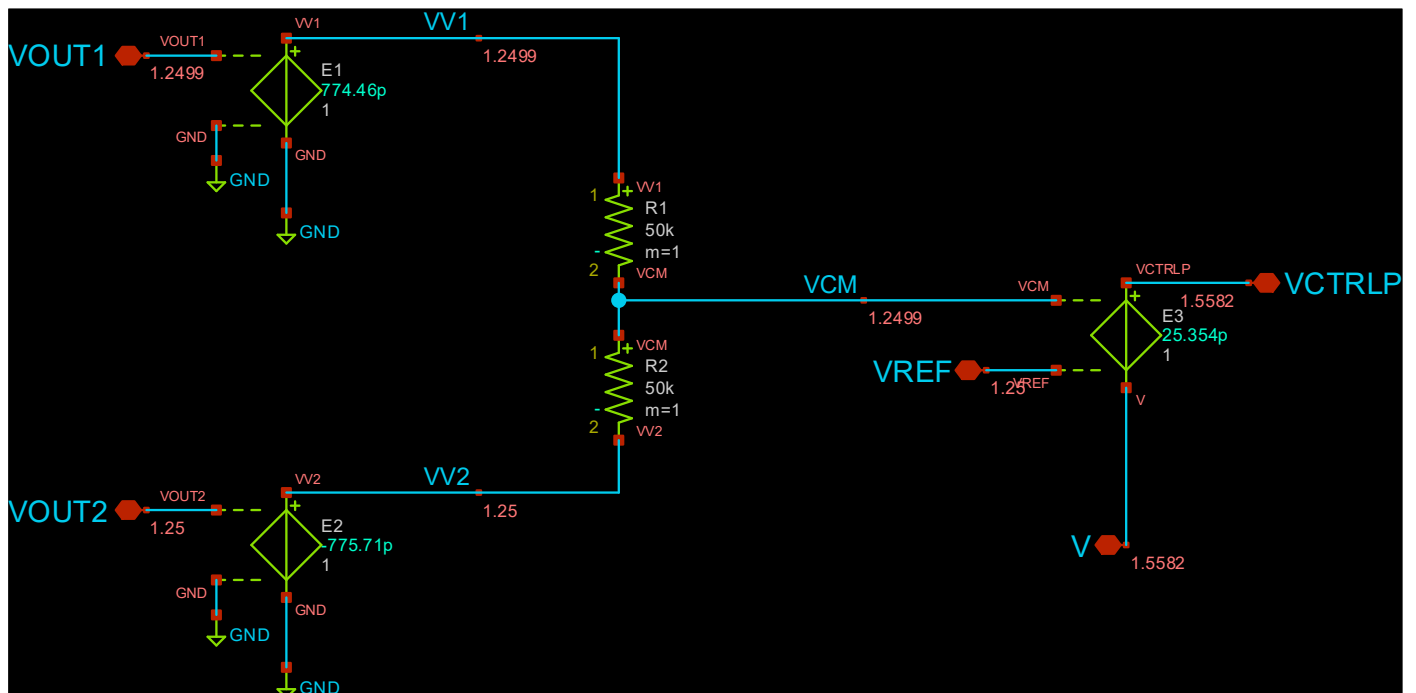
    let i = i + 1
  end

  * --- Print all results as a table ---
  print region_vec

endc
```

Index	region_vec
0	2.000000e+00
1	2.000000e+00
2	2.000000e+00
3	2.000000e+00
4	2.000000e+00
5	2.000000e+00
6	2.000000e+00
7	2.000000e+00
8	2.000000e+00
9	2.000000e+00
10	2.000000e+00
11	2.000000e+00
12	2.000000e+00
13	2.000000e+00
14	2.000000e+00
15	2.000000e+00
16	2.000000e+00
17	2.000000e+00

3) What are the differential input and output voltages of the error amplifier? What is the relation between them?



diff in = 5.666818e-05

differential input = 60uV and differential output will have the same value as the gain is 1

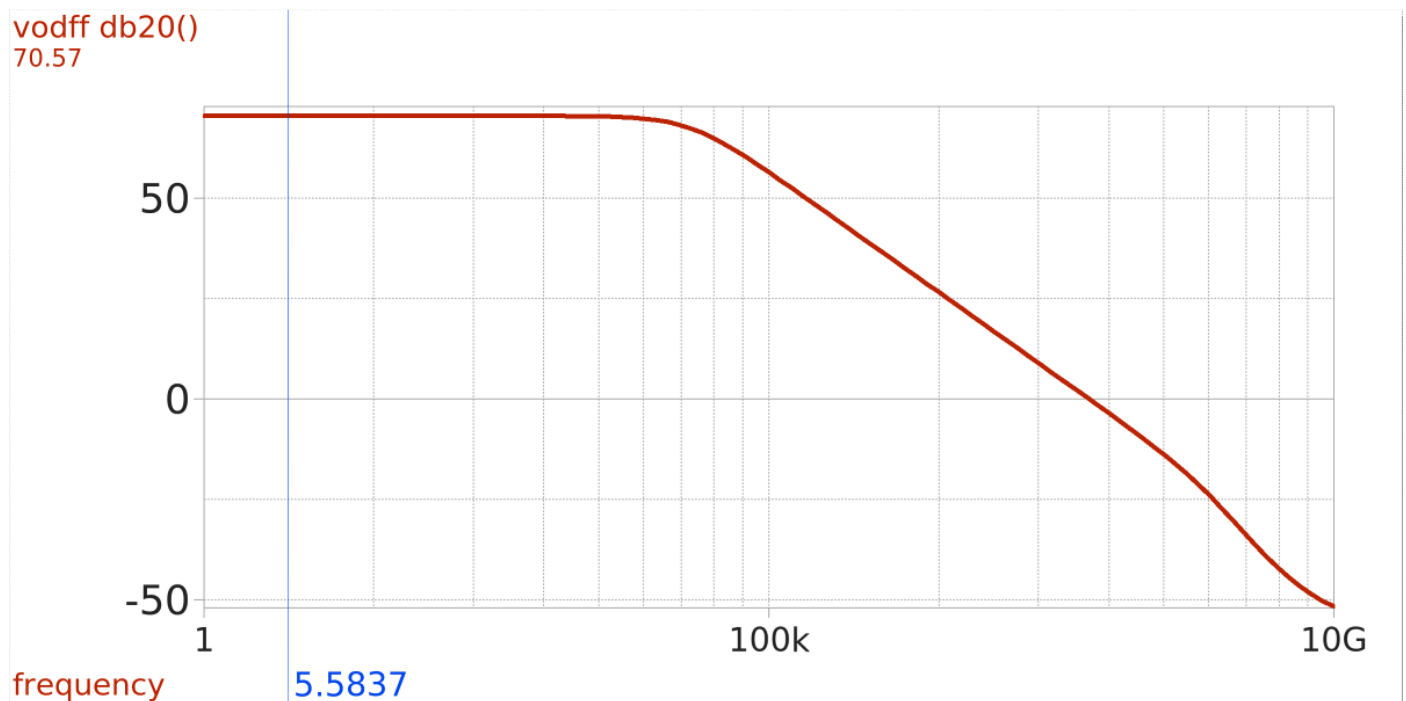
$$out = in * gain$$

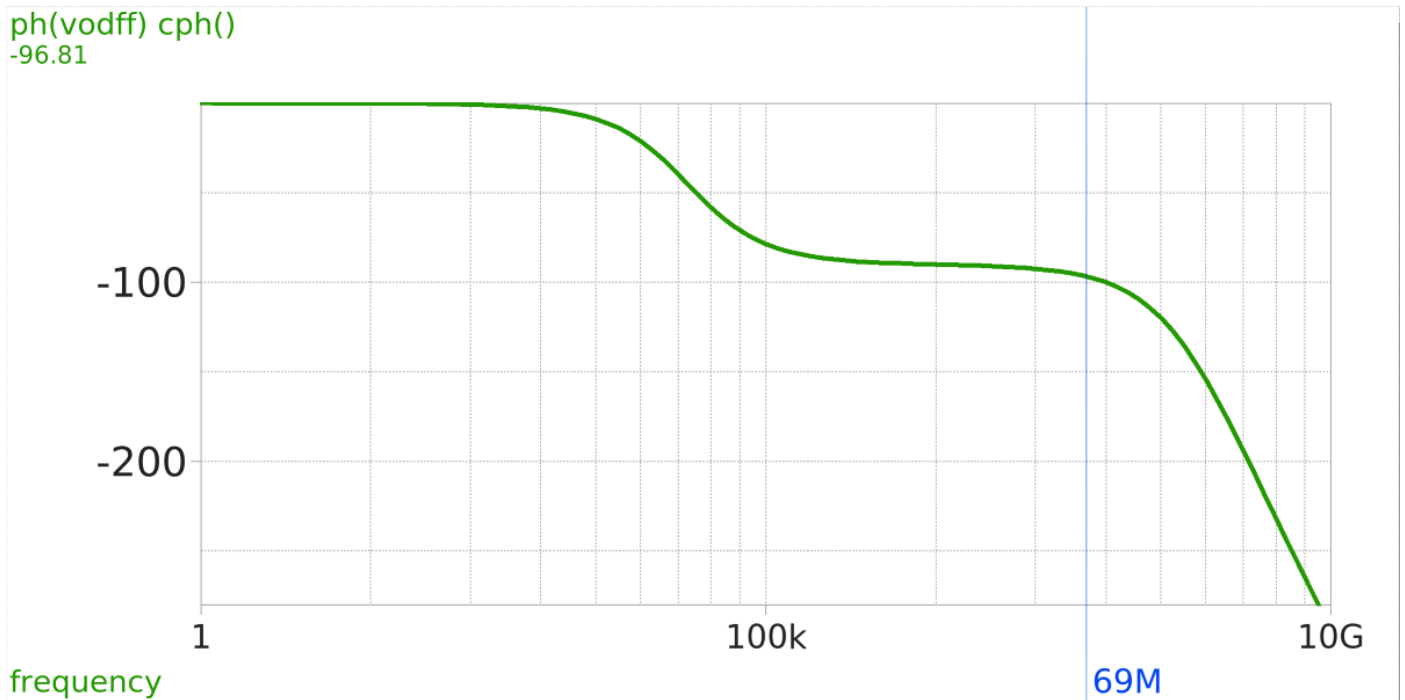
2) Diff small signal ccs

- 1) Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade).

```
s2
.control
save all
op
ac dec 10 1 10G
meas ac Gain MAX vmag(VODFF) FROM=1 TO=10G
let GAIN_db = db(GAIN)
print GAIN_db
let ff2=Gain*0.707
meas ac BW WHEN vmag(VODFF)=ff2 FALL =1
let GBW = Gain * BW
print GBW
meas ac GAIN_CROSSOVER_FREQ WHEN vmag(VODFF)=1
save v(VDIFF)
remzerovec
write folded_ota.raw
.endc
```

- 2) Plot diff gain (magnitude in dB and phase) vs frequency.





3) Calculate circuit parameters (DC gain, BW, GBW, UGF, and PM).

Phase margin = $180 - 96.81 = 83.19^\circ$

```
gain          = 3.377472e+03 at= 1.000000e+00
gain_db       = 7.057184e+01
bw            = 2.029831e+04
gbw           = 6.855697e+07
gain_crossover_freq = 6.906095e+07
```

4) Compare simulation results with hand calculations in a table (use SS parameters from OP simulation in your hand analysis).

$$R_{LFDN} = \frac{1}{1.195 \times 10^{-6}} \left(1 + (.000203 + 6.462 \times 10^{-6}) * \frac{1}{(4.314 + 8.848) \times 10^{-6}} \right) = 14.154 M\Omega$$

$$R_{LFDP} = \frac{1}{7.999 \times 10^{-7}} \left(1 + (.0002026 + 8.022 \times 10^{-5}) * \frac{1}{4.597 \times 10^{-6}} \right) = 78.163 M\Omega$$

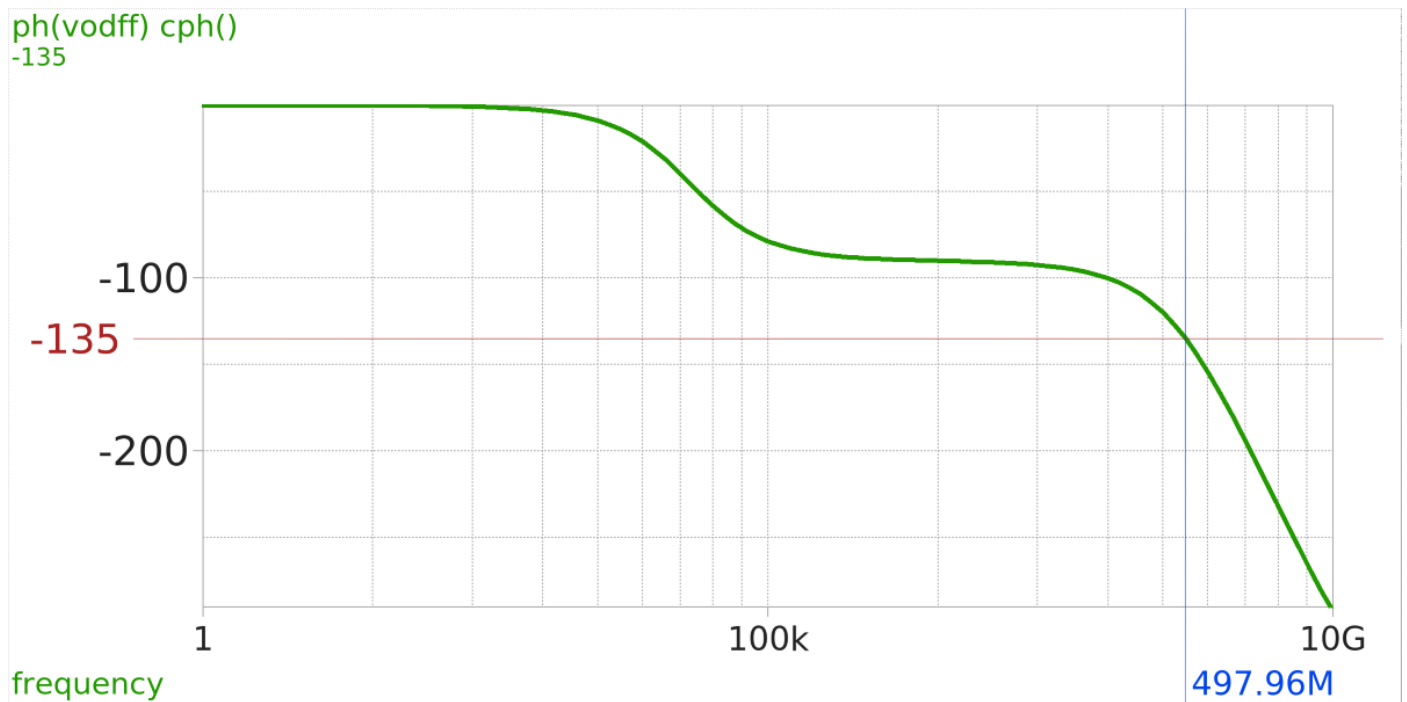
$$R_{out} = 11.984 M\Omega$$

$$gain = g_{m,input} * R_{out} = 242.8 * 10^{-6} * R_{out} = 2946$$

$$BW = \frac{1}{2\pi * (C_L + C_{dd}) * R_{out}} = 25.13 \text{ kHz}$$

$$GBW = 73.13 \text{ MHz}$$

24	cdd	NMOS	7.231f
24	cdd	PMOS	21.15f



If we assumed our system has only two poles the second pole is at phase = -135°

$$pm = 90 - \tan^{-1} \left(\frac{69}{497.96} \right) = 82.1^\circ$$

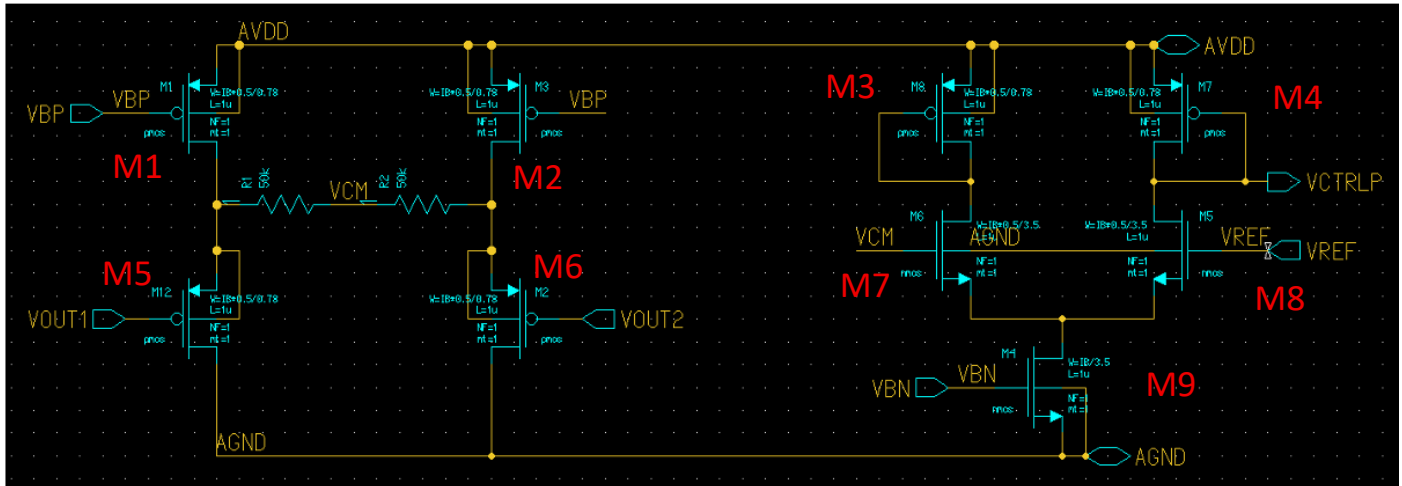
Since our result is close to the analytical results then our assumption is valid

	Simulation	Analytical
Gain	3377	2946
BW	20 KHz	25.13 KHz
GBW	68.6 MHz	73.13 MHz
PM	83.19°	82.1°

PART 4: Open-Loop OTA Simulation (Actual CMFB)

1) Design

i selected the current in each branch to be 8uA



- M1,2,3 and 4 should match the tail current source so they have the same L (1u), (W will be 15.61u)
- M9 should match the NMOS sources (W will be 7.29u)

Now we only have four transistors to size (You may assume $L = 1\mu\text{m}$ and $gm/ID = 15$)

1) M5,6

when using $L = 1\mu\text{m}$ the width is large and the parasitics so the CM phase margin was 68 (not good) so I used min L instead

LUT Settings

ID	8u
gm/ID	15
L	min
VDS	1
VSB	0
Stack	1

Results:

	Name	TT-27.0
1	ID	8u
2	IG	N/A
3	L	280n
4	W	10.08u
5	VGS	788.3m
6	VDS	1
7	VSB	0
8	gm/ID	14.72
9	Vstar	135.9m

final sizing

ID	8u
gm/ID	15
L	1u
VDS	1
VSB	0
Stack	1

Results:

	Name	TT-27.0
1	ID	8u
2	IG	N/A
3	L	1u
4	W	51.21u
5	VGS	846.3m
6	VDS	1
7	VSB	0
8	gm/ID	14.85
9	Vstar	134.7m

first sizing

2) M7,8

Using small L to improve the CM phase margin

For the sensing resistance I set it to 25K instead of 50K also to improve the CM phase

ID	8u	?
gm/ID	15	?
L	300n	?
VDS	0.8	?
VSB	0.8	?
Stack	1	?

Results:

	Name	TT-27.0
1	ID	8u
2	IG	N/A
3	L	300n
4	W	4.04u
5	VGS	838.2m
6	VDS	800m
7	VSB	800m
8	gm/ID	14.72
9	Vstar	135.9m

3) Vref selection

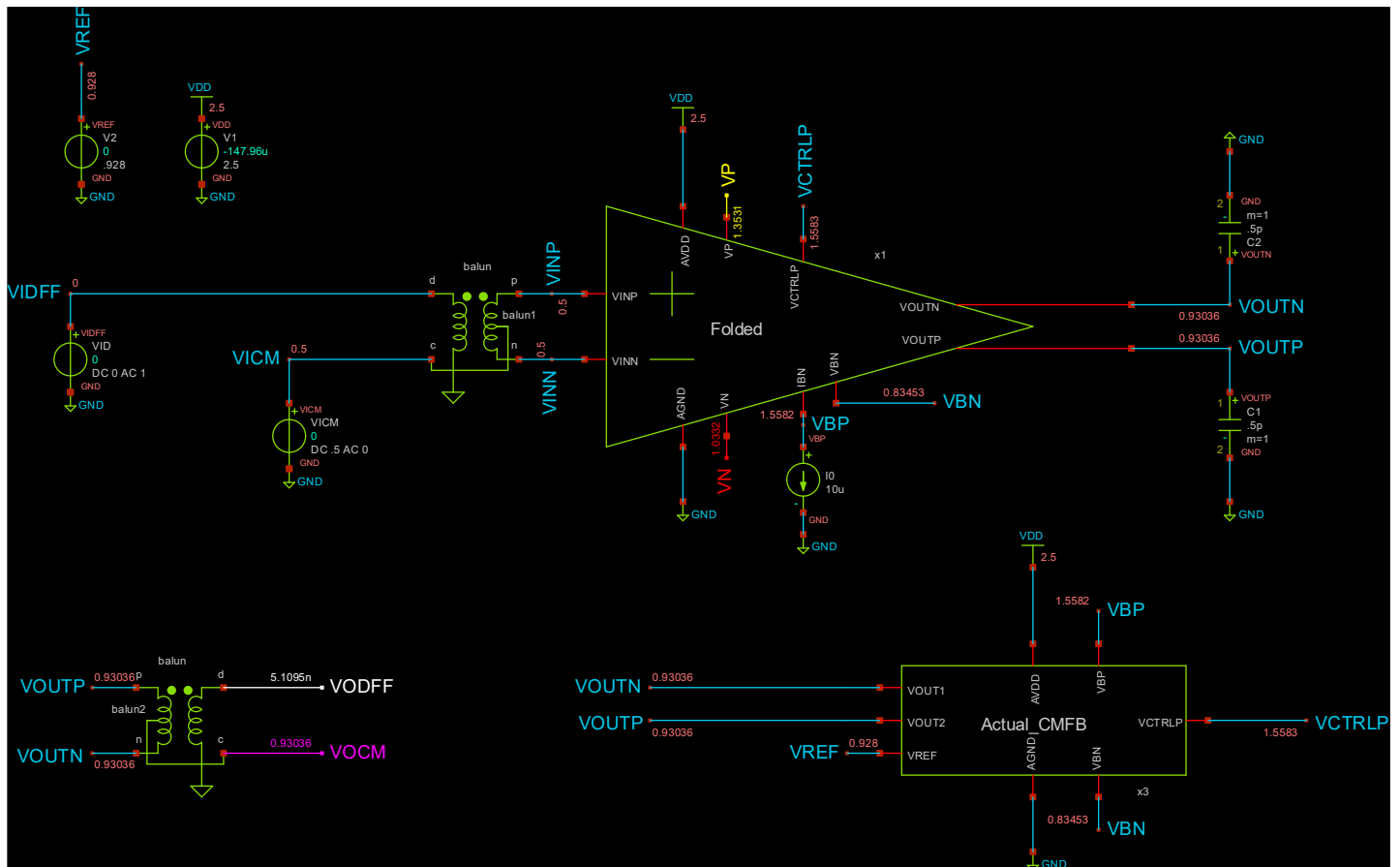
$$\text{Max} = VDD - V^* - |VGSP| = 2.5 - .778 - \frac{2}{10} = 1.522V$$

$$\text{Min} = 2 V^* = .33V$$

$$\text{Average} = V_{\text{ref}} = .9276V \approx 928mV$$

We need to apply a shift to Vref.

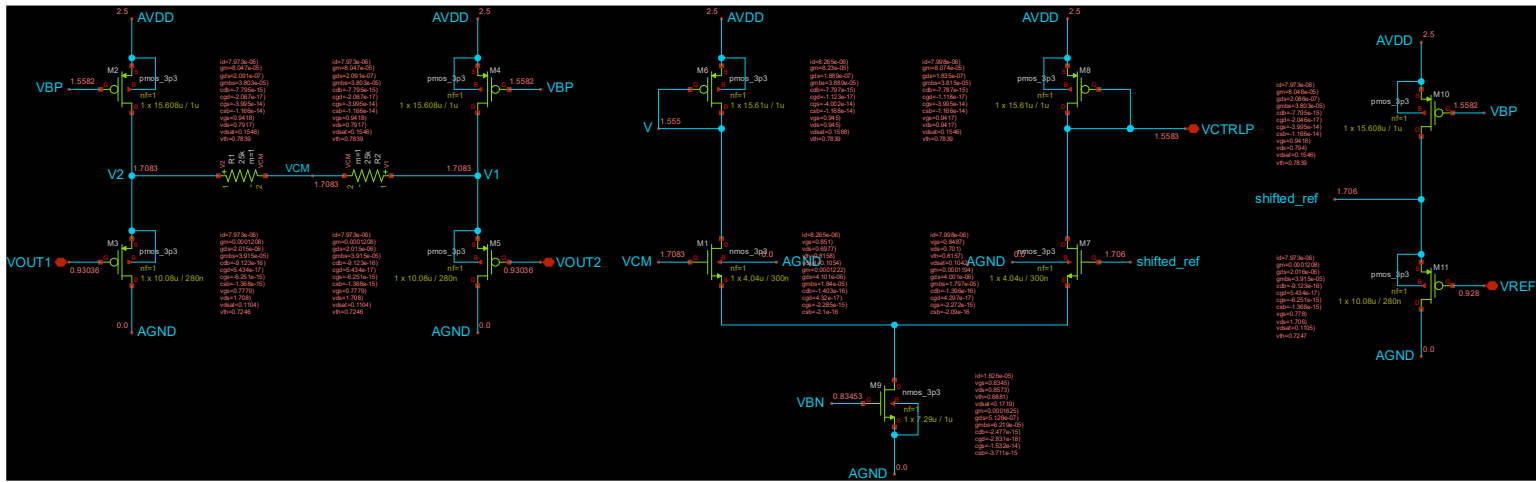
2) OP



1) What is the CM level at the OTA output? Why?

CM = 924.25mV, which is approximately Vref

- The CMFB senses the average of the outputs.
- It compares this average to Vref.
- The error amplifier drives the OTA bias currents so that $V_{CM_out} \rightarrow V_{ref}$
- Thus, the output CM is **forced** to follow Vref



2) What are the differential input and output voltages of the error amplifier? What is the relation between them?

$$\text{differential input} = 1.7083 - 1.706 = 2.3\text{mV}$$

$$\text{differential output} = 1.5583 - 1.555 = 3.3\text{mV}$$

$$\text{error amplifier gain} = \frac{g_{m7}}{g_{m8}} = \frac{.0001194}{8.074 \times 10^{-5}} = 1.4788$$

The relationship between them is the error amplifier gain $\frac{out}{in} = 1.44$

Index	region_vec
0	2.000000e+00
1	2.000000e+00
2	2.000000e+00
3	2.000000e+00
4	2.000000e+00
5	2.000000e+00
6	2.000000e+00
7	2.000000e+00
8	2.000000e+00
9	2.000000e+00
10	2.000000e+00

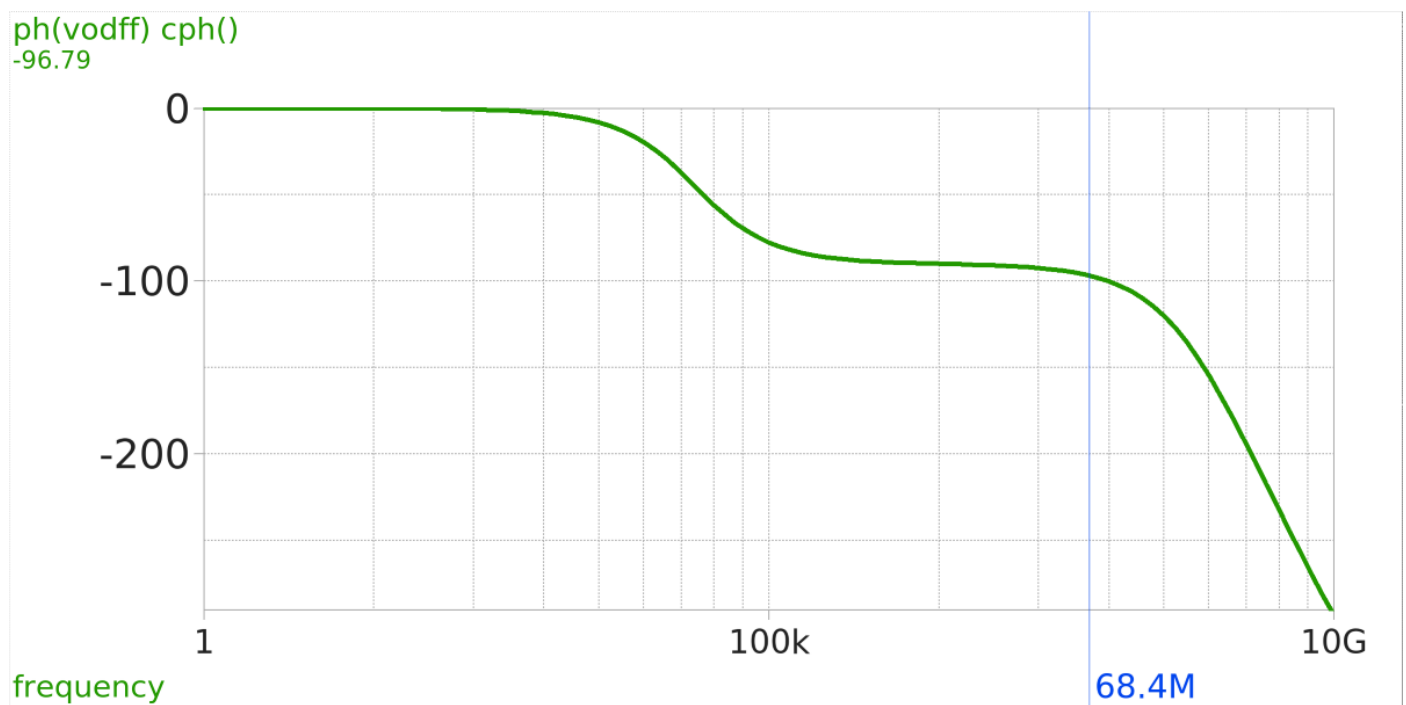
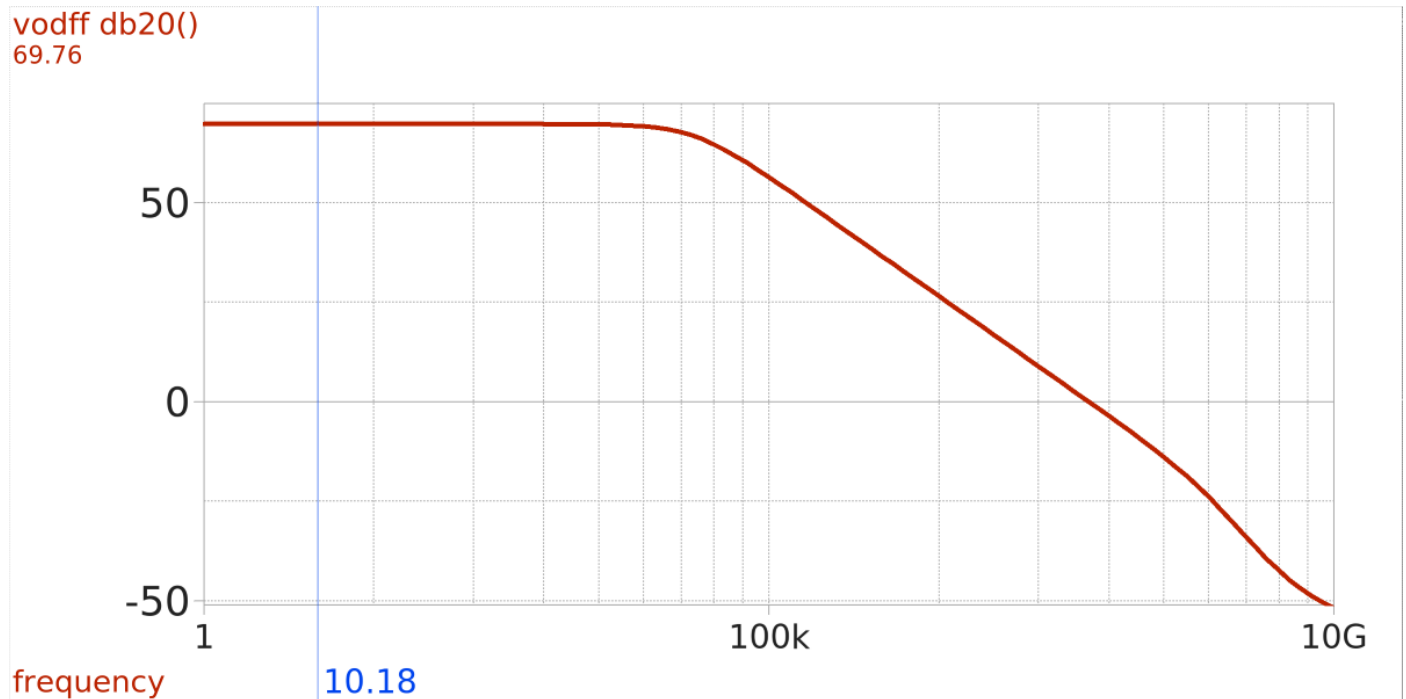
Region for CMFB circuit

Index	region_vec
0	2.000000e+00
1	2.000000e+00
2	2.000000e+00
3	2.000000e+00
4	2.000000e+00
5	2.000000e+00
6	2.000000e+00
7	2.000000e+00
8	2.000000e+00
9	2.000000e+00
10	2.000000e+00
11	2.000000e+00
12	2.000000e+00
13	2.000000e+00
14	2.000000e+00
15	2.000000e+00
16	2.000000e+00
17	2.000000e+00

Region for folded OTA circuit

3) Diff small signal ccs

1) Plot diff gain (magnitude in dB and phase) vs frequency.



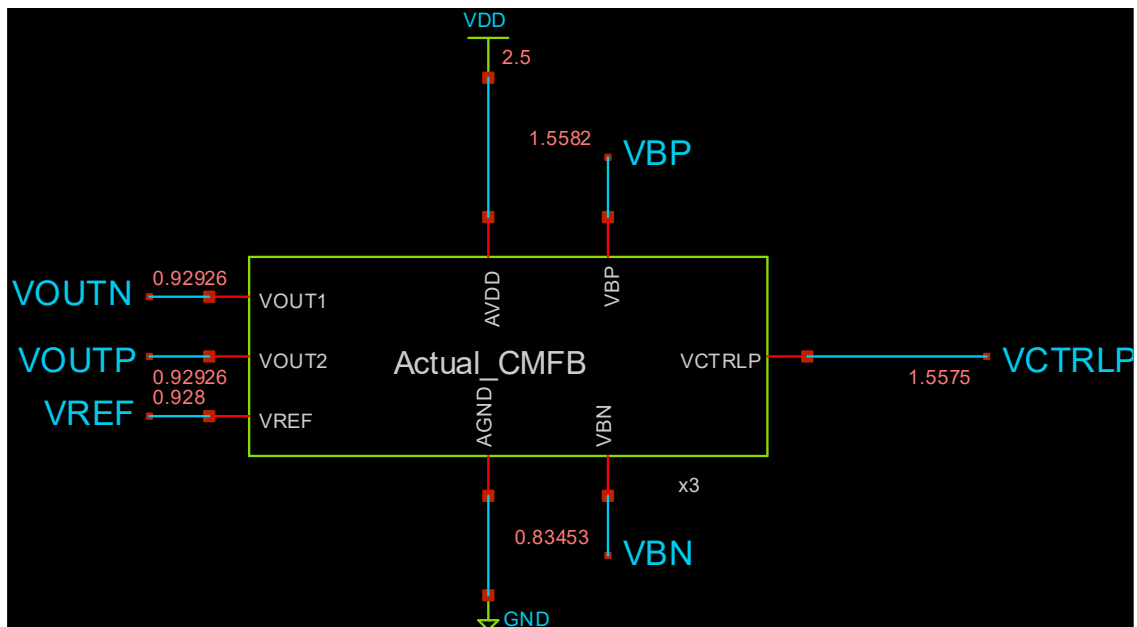
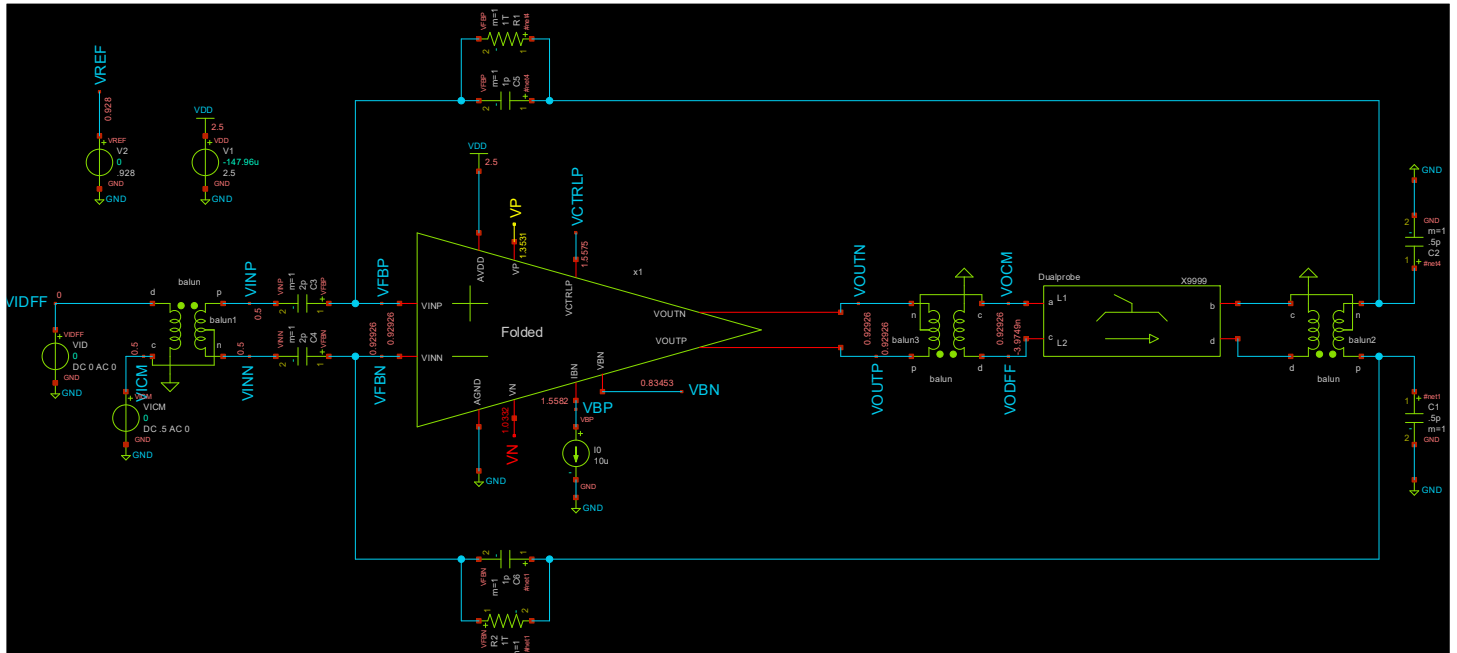
2) Calculate circuit parameters (DC gain, BW, GBW, UGF, and PM).

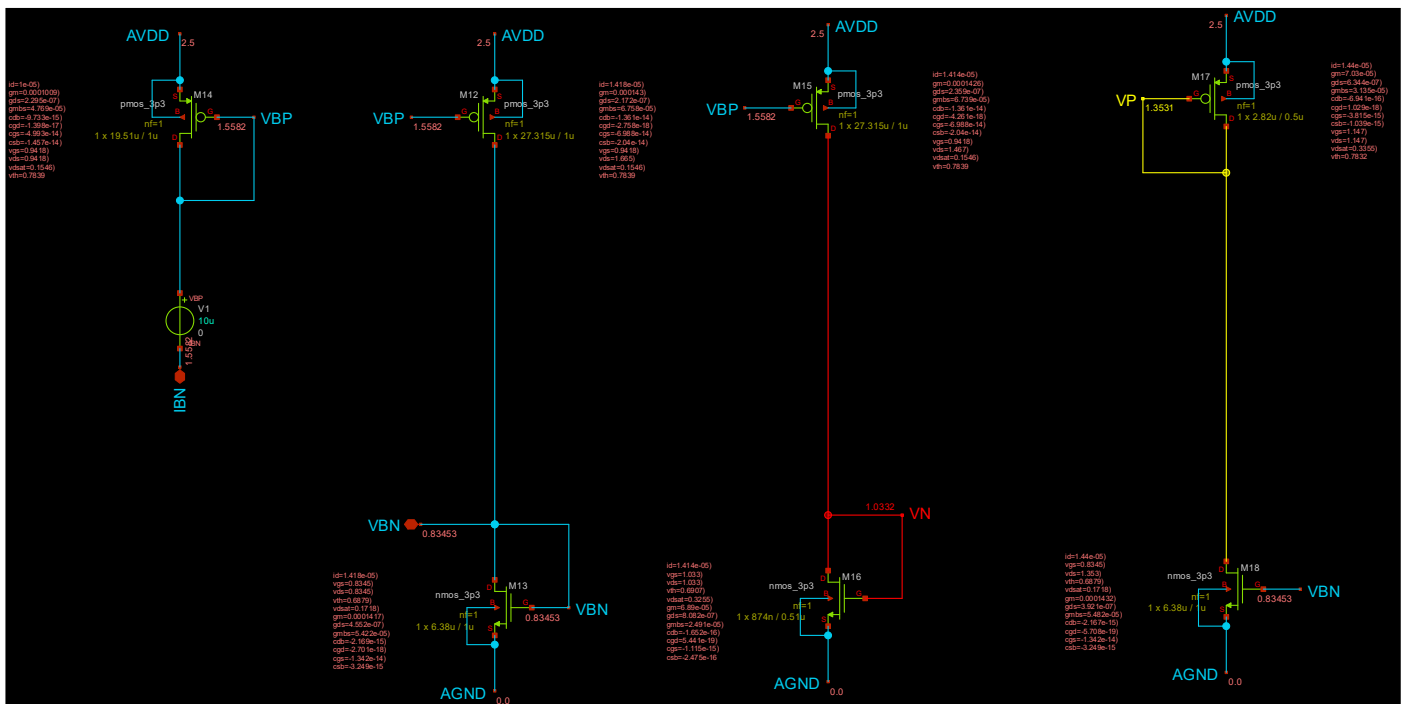
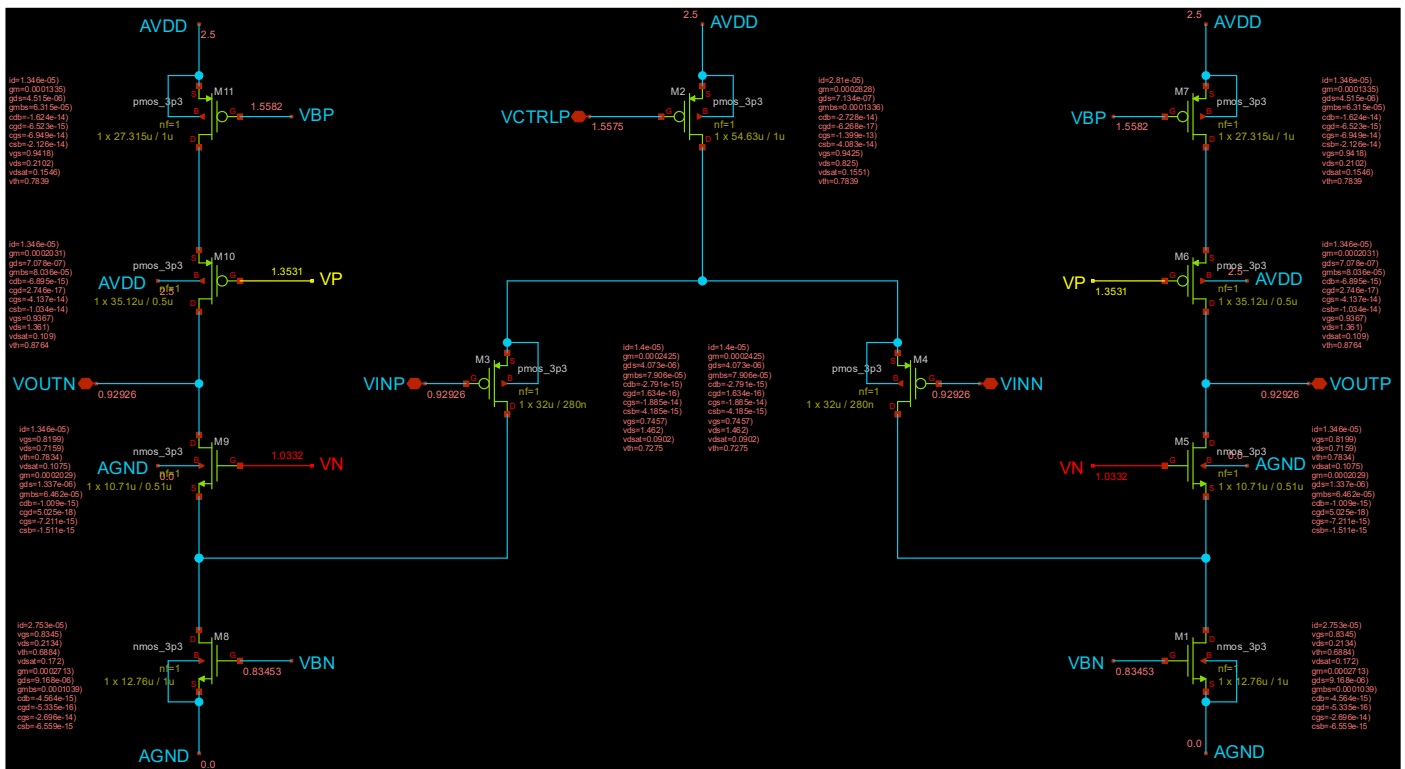
$$pm = 180 - 96.79 = 83.19^\circ$$

```
gain          = 3.077037e+03
gain_db       = 6.976265e+01
bw            = 2.213341e+04
gbw           = 6.810532e+07
gain_crossover_freq = 6.840515e+07
```

PART 5: Closed Loop Simulation (AC and STB Analysis)

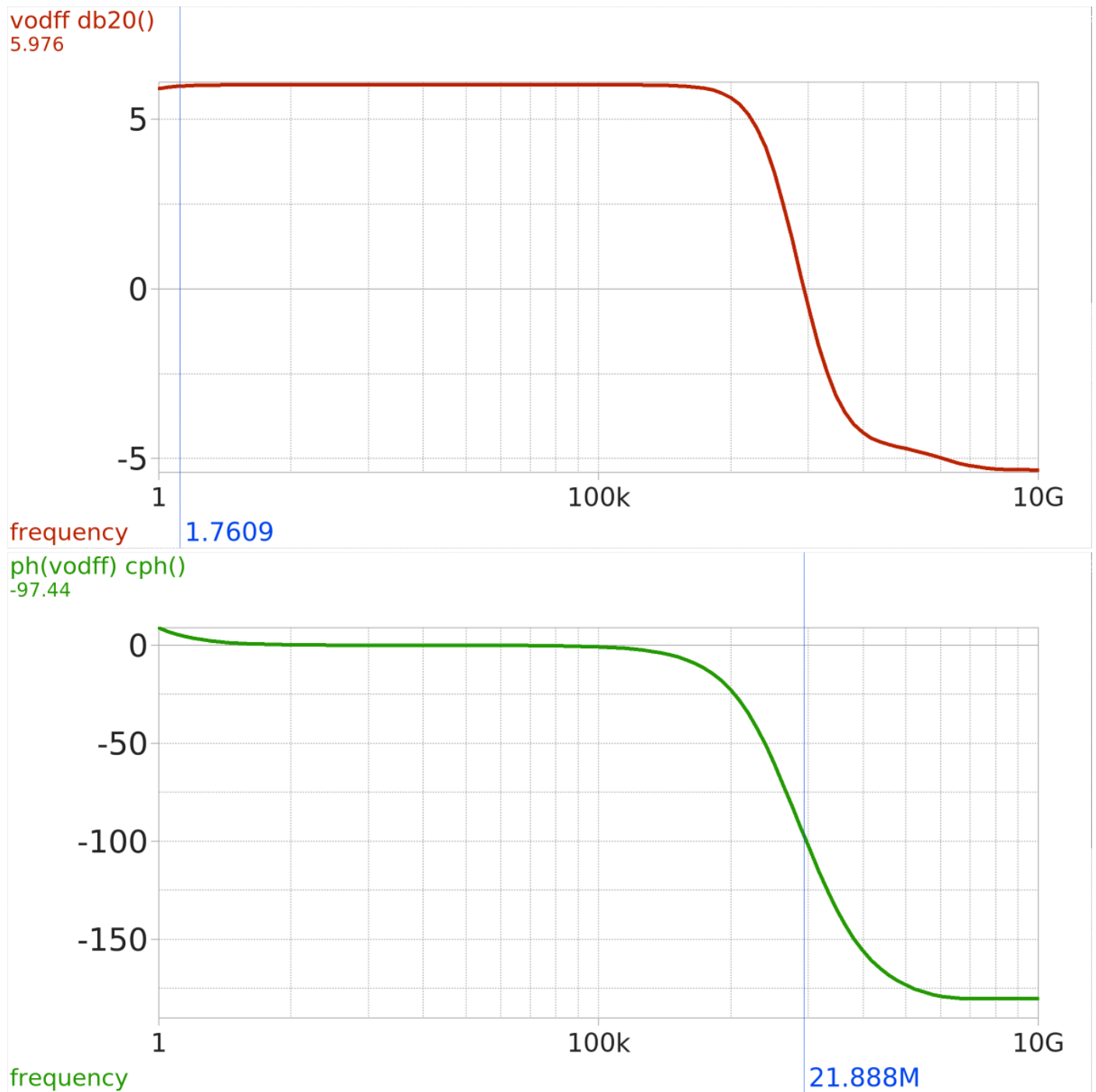
1) OP





2) Differential closed-loop response:

1) Plot VODIFF vs frequency



2) calculate circuit parameters (DC gain, CL BW, CL GBW)

```
gain          = 1.998017e+00  
gain_db       = 6.011984e+00  
bw            = 1.121344e+07  
gbw           = 2.240464e+07  
gain_crossover_freq = 2.188870e+07
```

3) Differential and CMFB loops stability (STB analysis)

1) Run STB analysis

```
s8

.func tian_loop_1() {1/(1-1/(2*(ac1.I(v.X9999.Vi)*ac2.V(X9999.x)-ac1.V(X9999.x)*ac2.I(v.X9999.Vi))+ac1.V(X9999.x)+a
.func tian_loop_2() {1/(1-1/(2*(ac3.I(v.X9999.Vi1)*ac4.V(X9999.y)-ac3.V(X9999.y)*ac4.I(v.X9999.Vi1))+ac3.V(X9999.y)

*-----
* Probes
*-----
.save V(X9999.x) I(v.X9999.Vi)
.save V(X9999.y) I(v.X9999.Vi1)

*-----
* CONTROL BLOCK
*-----
.control

    set filetype=ascii
    set num_threads=8
    set color0=white
    set color1=black
    set xbrushwidth =3
    unset askquit

    optran 0 0 0 100n 4u 0
    op
    write stb.raw
    set appendwrite

*-----
* First Loop STB analysis
*-----
* Set voltage AC to 1
alter i.X9999.Ii1 acmag=0
alter v.X9999.Vi1 acmag=0
alter v.X9999.Vi acmag=1
ac dec 50 100 10G

* Set Current to 1
alter i.X9999.Ii acmag=1
alter v.X9999.Vi acmag=0

ac dec 50 100 10G

*-----
* Second Loop STB analysis
*-----
* Set voltage AC to 1
alter i.X9999.Ii acmag=0
alter v.X9999.Vi acmag=0
alter v.X9999.Vi1 acmag=1

print loop1_LG
ac dec 50 100 10G

* Set Current to 1
alter i.X9999.Ii1 acmag=1
alter v.X9999.Vi1 acmag=0
ac dec 50 100 10G

let tian_signal_1 = tian_loop_1()

let loop_L1_gain_db = db(tian_signal_1)
let loop_L1_gain_ph = 180*cph(tian_signal_1)/pi

*plot loop_L1_gain_db
*plot loop_L1_gain_ph

save tian_signal_1

meas ac Gain_1 MAX vmag(tian_signal_1) FROM=1 TO=10G
let LG_L1 = 20*log10(Gain_1)
meas ac L1_GAIN_CROSSOVER_FREQ WHEN loop_L1_gain_db=0
meas ac L1_phaseatzerogain FIND loop_L1_gain_ph AT=L1_GAIN_CROSSOVER_FREQ
*let L1_PM = abs(180 - L1_phaseatzerogain)
*print L1_PM

let loop1_LG = LG_L1
let loop1_GX = L1_GAIN_CROSSOVER_FREQ
let loop1_PM = L1_phaseatzerogain

save loop1_LG
save loop1_GX
save loop1_PM

let tian_signal_2 = tian_loop_2()

let loop_L2_gain_db = db(tian_signal_2)
let loop_L2_gain_ph = 180*cph(tian_signal_2)/pi

*plot loop_L2_gain_db
*plot loop_L2_gain_ph

save tian_signal_2

meas ac Gain_2 MAX vmag(tian_signal_2) FROM=1 TO=10G
let LG_L2 = 20*log10(Gain_2)
meas ac L2_GAIN_CROSSOVER_FREQ WHEN loop_L2_gain_db=0
meas ac L2_phaseatzerogain FIND loop_L2_gain_ph AT=L2_GAIN_CROSSOVER_FREQ
*let L2_PM = abs(180 - L2_phaseatzerogain)
*print L2_PM

let loop2_LG = LG_L2
let loop2_GX = L2_GAIN_CROSSOVER_FREQ
let loop2_PM = L2_phaseatzerogain

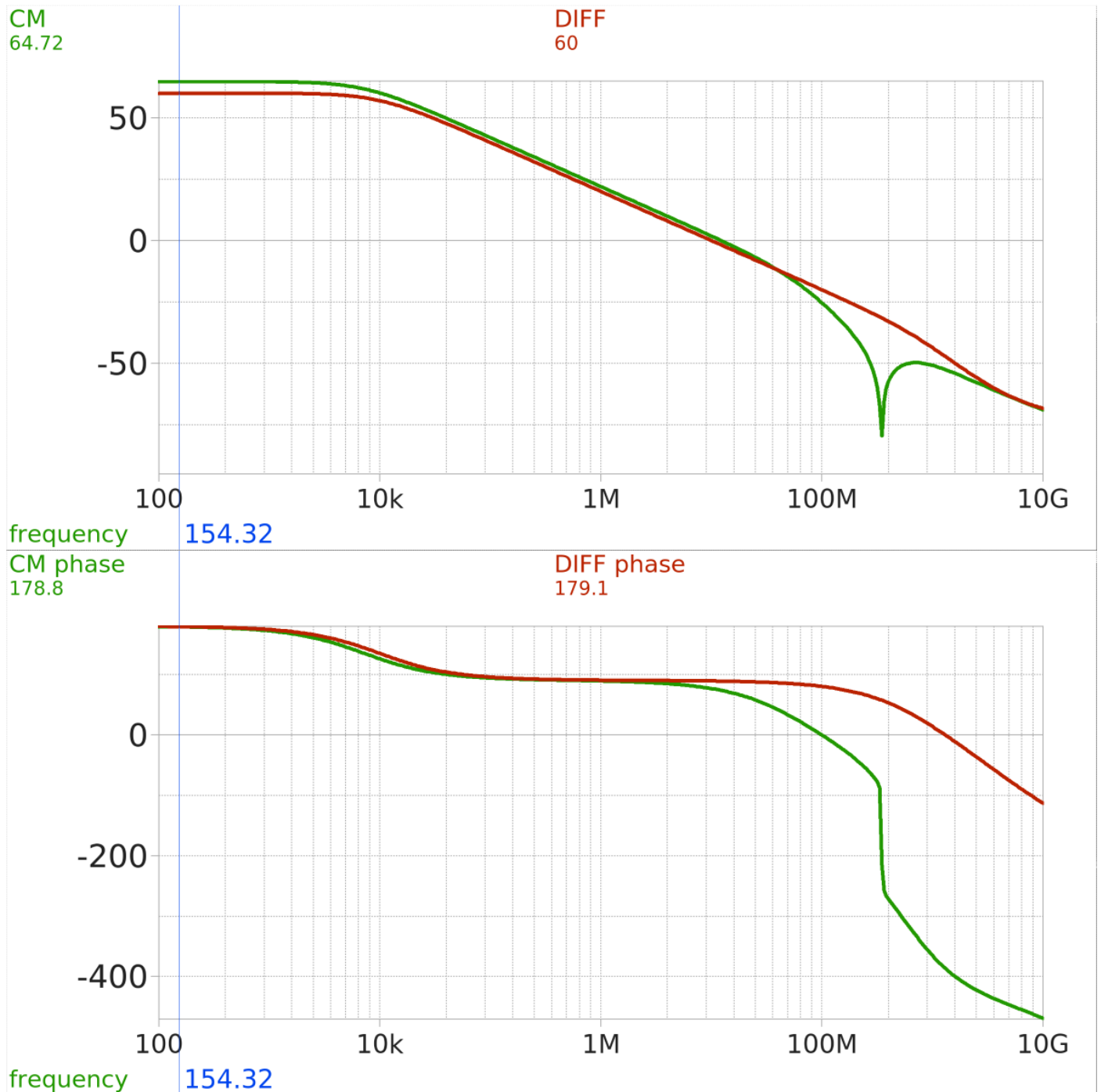
save loop2_LG
save loop2_GX
save loop2_PM

echo -----
print loop1_LG loop1_GX loop1_PM
echo -----
print loop2_LG loop2_GX loop2_PM
echo -----

write Dstb.raw tian_signal_1 tian_signal_2

.endc
```

2) Plot loop gain in dB and phase vs frequency for the two simulations overlaid.



loop1 is CM and loop2 is diff

```
loop1_lg = 6.475826e+01
loop1_gx = 1.221570e+07
loop1_pm = 7.380721e+01
-----
loop2_lg = 5.999955e+01
loop2_gx = 9.983662e+06
loop2_pm = 8.907843e+01
```

3) Compare GBW and PM of diff and CM loops. Comment.

	DIFF	CM
GBW	9.98 MHz	12.2 MHz
PM	89°	73.8°

The differential loop shows a GBW of about 10 MHz with a very high phase margin ($\approx 89^\circ$ overdamped), indicating excellent stability. The common-mode loop has a slightly higher GBW (≈ 12.2 MHz), but a lower phase margin ($\approx 74^\circ$). This is expected, since the CMFB loop is designed with less margin to avoid slowing down common-mode regulation. Both loops are stable, but the differential loop is more robust, while the CMFB loop trades some phase margin for faster common-mode control.

4) Compare DC LG and GBW of the diff loop with those obtained from open-loop simulation. Comment

	LG	OL
Gain	60 dB	69.8 dB
GBW	9.98 MHz	68.1 MHz

The LG is 1000 and OL gain is 3077 and as we know:

$$LG = A_{OL} * \beta = 3077 * \frac{1}{3} = 1026$$

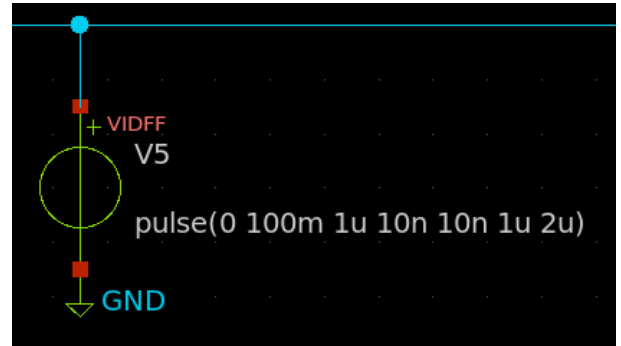
The analytical value is a bit higher because beta is lower than $1/3$ due to the parasitics, but since the result is close the approximation is valid

The LG GBW is much smaller than the OL GBW due to loading effect and parasitics.

PART 6: Closed Loop Simulation (Transient Analysis)

1) Differential and CMFB loops stability

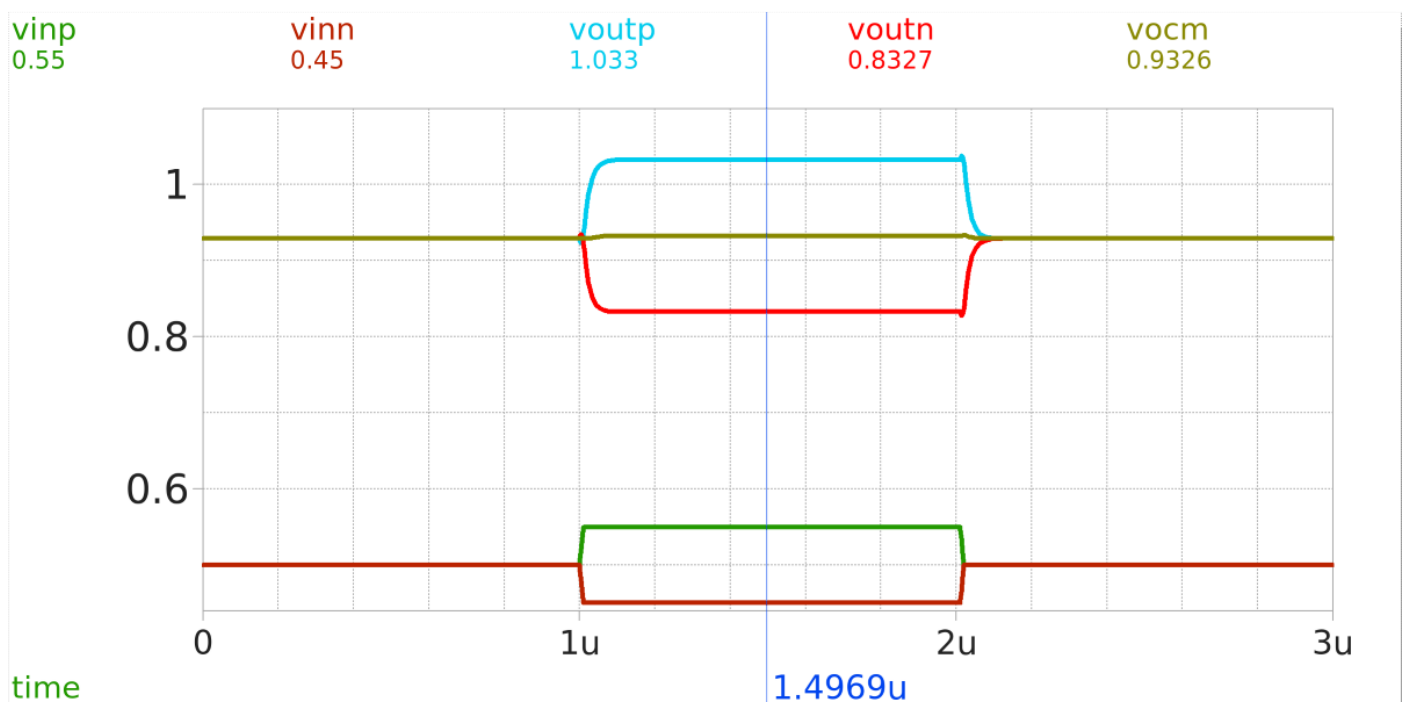
- 1) Apply a differential input pulse (initial value = 0, pulse value = 100mV, delay = 1 μ s, period = 2 μ s, pulse width = 1 μ s, rise = fall = 10ns).



- 2) Run transient analysis for 3 μ s with 10ns max step.

```
s5
.control
save all
tran 10n 3u
write folded_loop.raw
.endc
```

- 3) Plot the transient signals at VINP, VINN, VOUTP, VOUTN, and VOCM overlaid in the same figure.

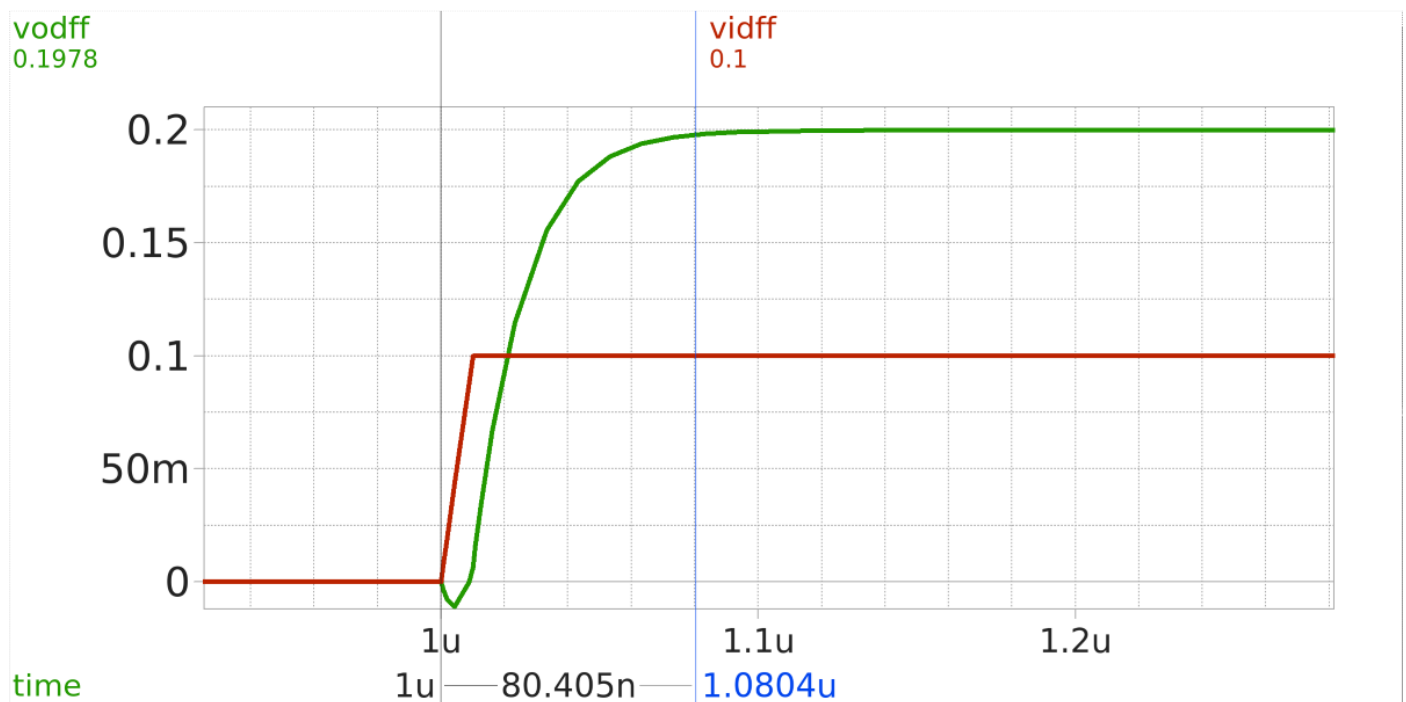


4) Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

No, there is no differential ringing as we have a high PM (overdamped response), for the CM there is no ringing also but there is an overshoot due to the response is underdamped.

both loops are stable with adequate PM

5) Calculate the 1% settling time and compare it to the required specification. If the specification is not satisfied, what design changes could be a possible solution?



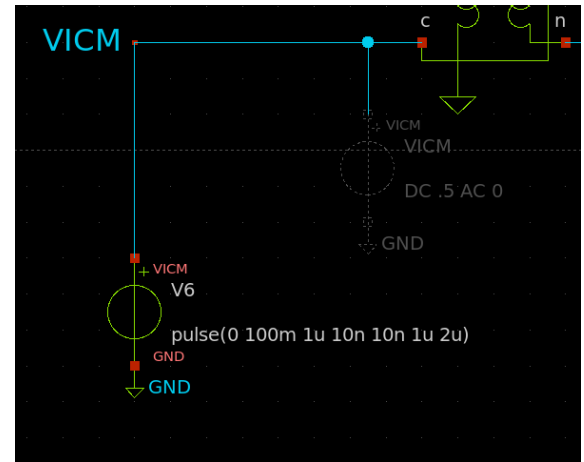
$$1\% \text{ error value} = .99 * \text{final value} = .99 * .1998 = .1978$$

	Achieved	Required
Settling time	80.4 ns	<100ns

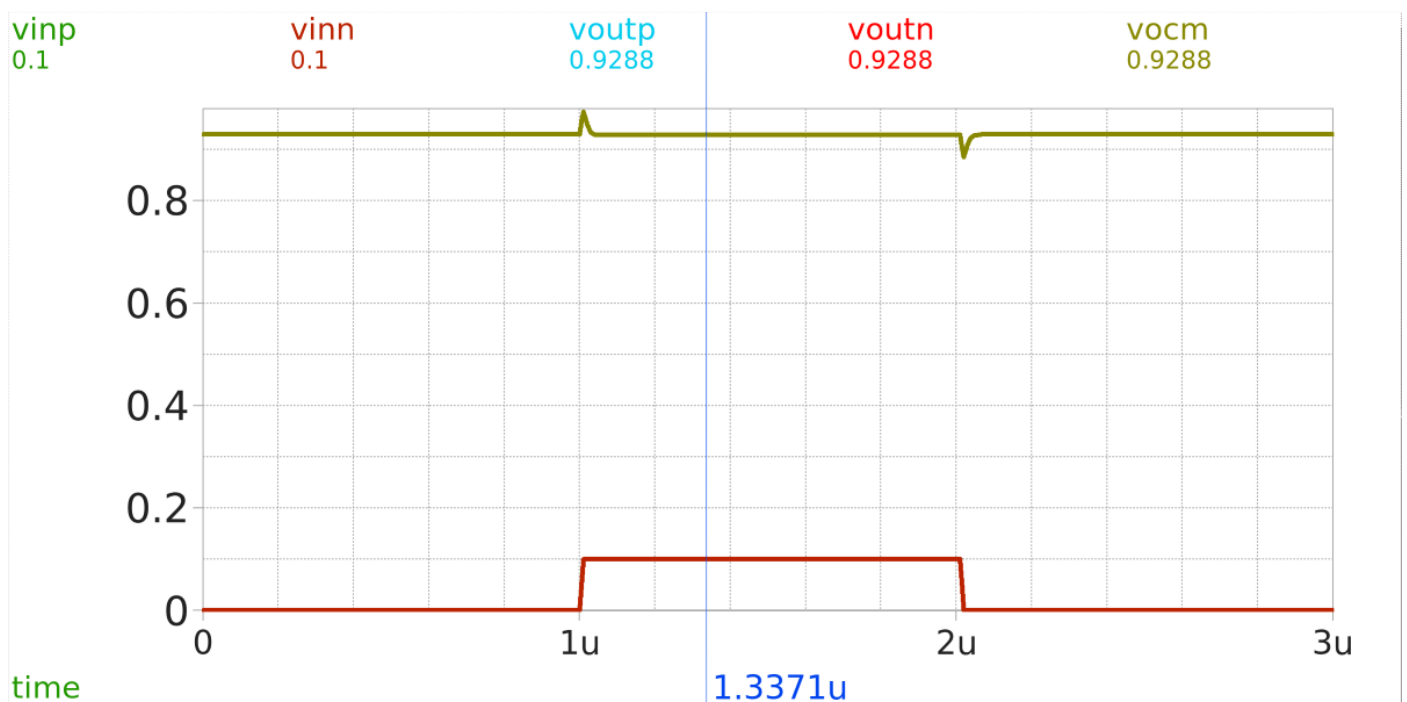
If it wasn't achieved we should increase our GBW (input pair width)

2) Differential and CMFB loops stability (transient analysis): CM input pulse

- 1) Set differential input to zero and apply the same previous pulse at the balun CM input.



- 2) Plot the transient signals at VINP, VINN, VOUTP, VOUTN, and VOCM overlaid in the same figure.



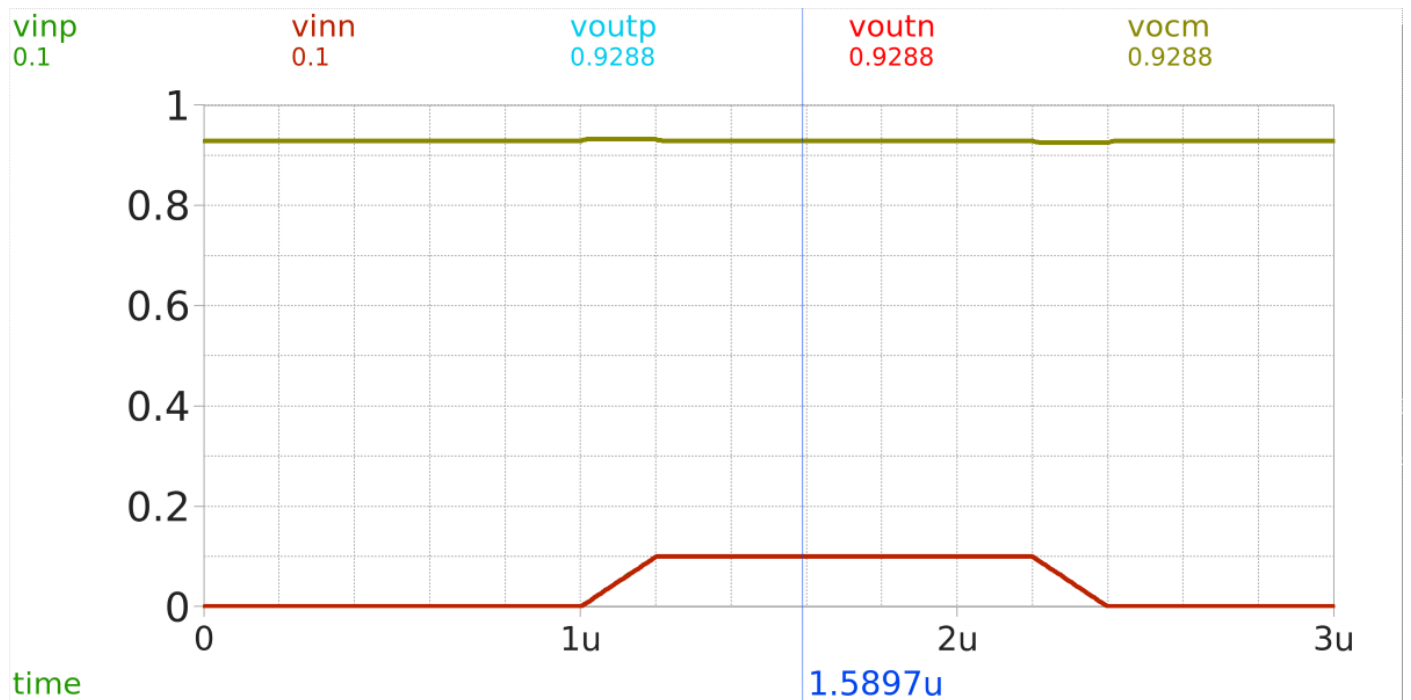
- 3) Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

No, there is no differential ringing as we have a high PM (overdamped response), for the CM there is no ringing also but there is an overshoot due to the response is underdamped.

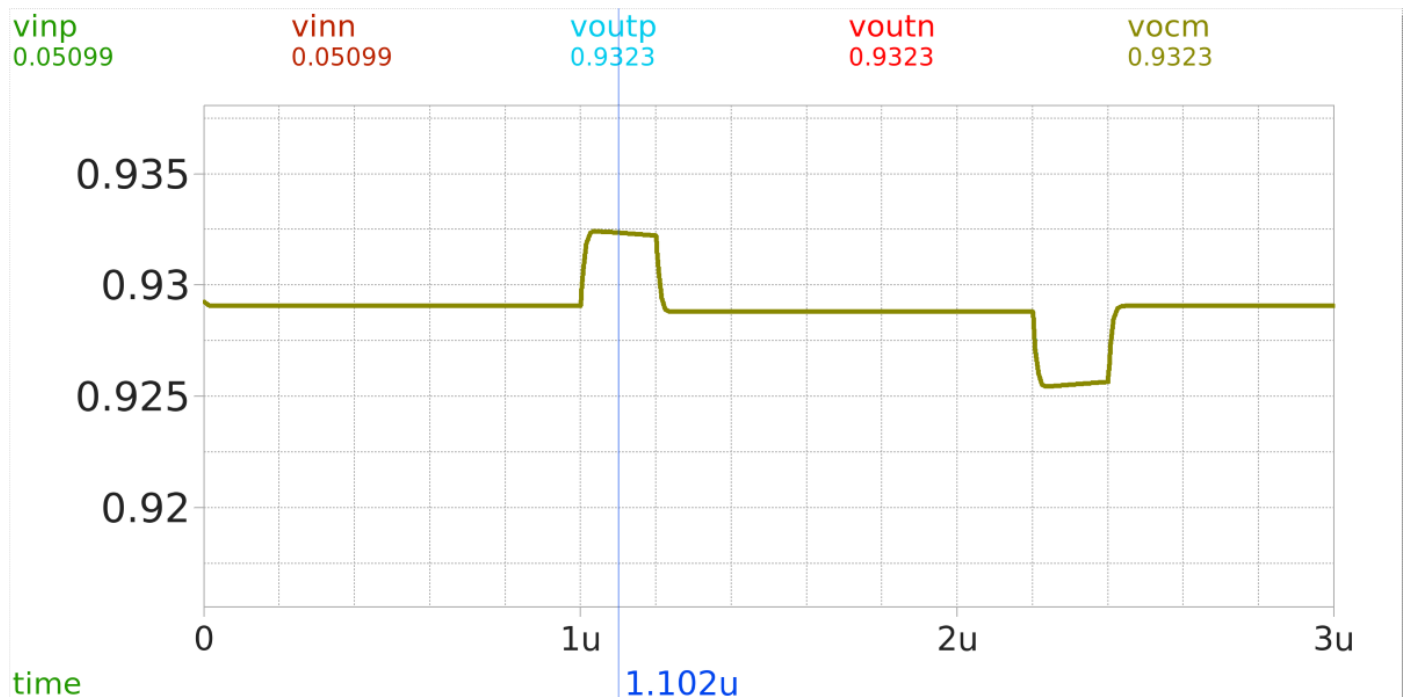
In nutshell both loops are stable with adequate PM

The peak value here has two components the CM overshoot and the caps effects when a sudden change in the volt across them happens

To verify our point let's increase the rise and fall times of the input pulse

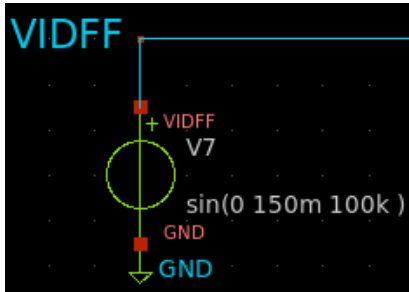


As we see the peak values are much smaller now as it's main component is eliminated (the sudden change across caps)



3) Output swing

- 1) Apply a differential sinusoidal input with freq = 100kHz and amplitude = 150mV.
- 2) Run transient analysis for three periods (30us) with 0.1us max time step.

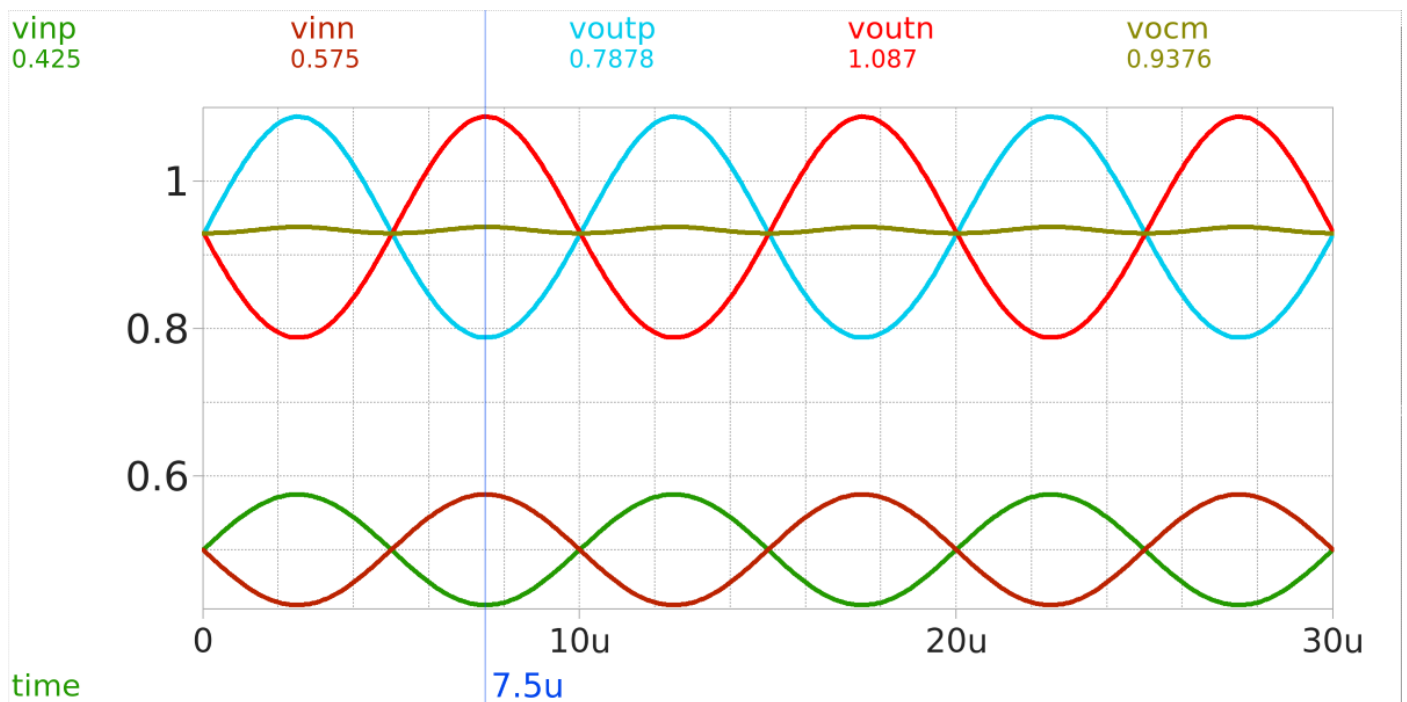


s6

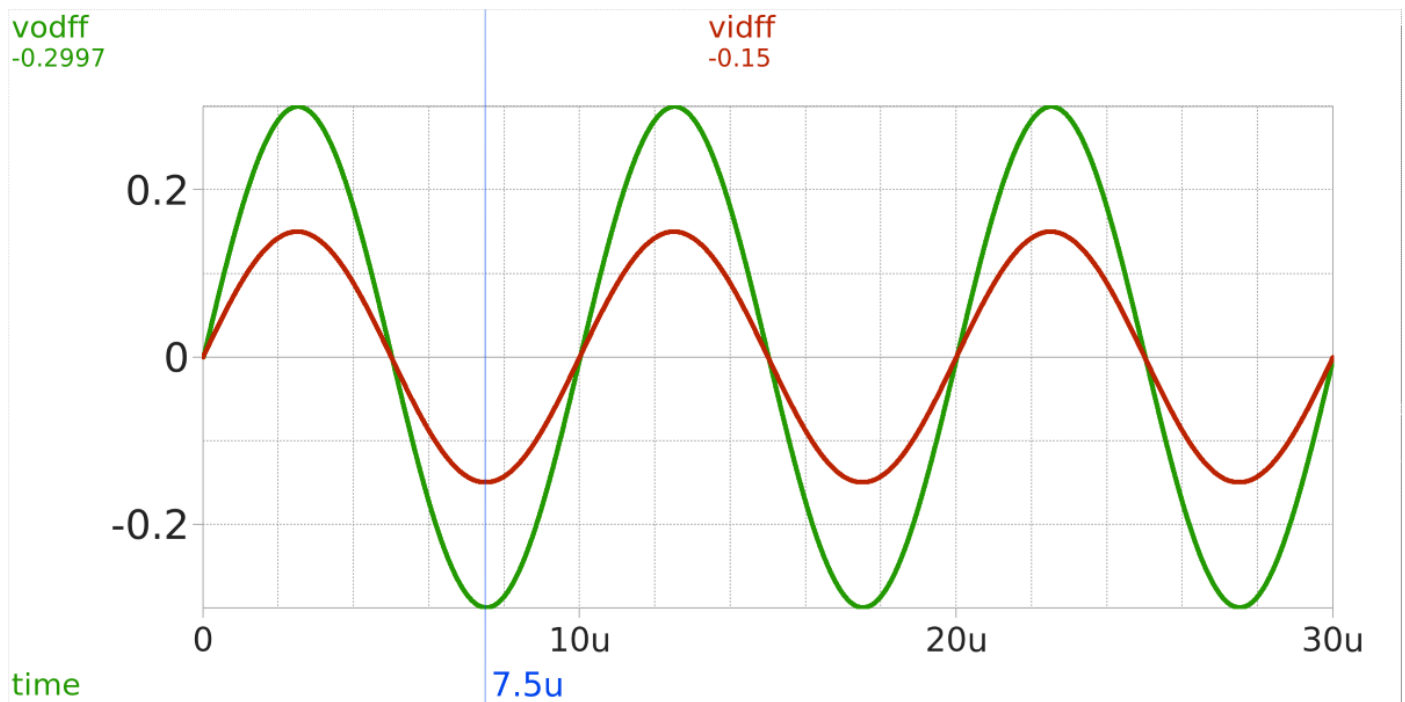
```
.control
save all
tran .1n 30u
meas tran vinmax max v(VIDFF)
meas tran vinmin min v(VIDFF)
meas tran voutmax max v(VODFF)
meas tran voutmin min v(VODFF)

let vin_pp = vinmax - vinmin
let vout_pp = voutmax - voutmin
let gain = vout_pp / vin_pp
print vin_pp vout_pp gain
write folded_loop.raw
.endc
```

- 3) Plot the transient signals at VINP, VINN, VOUTP, VOUTN, and VOCM overlaid in the same figure.



4) Plot the transient signals at VIDIFF and VODIFF overlaid in the same figure.

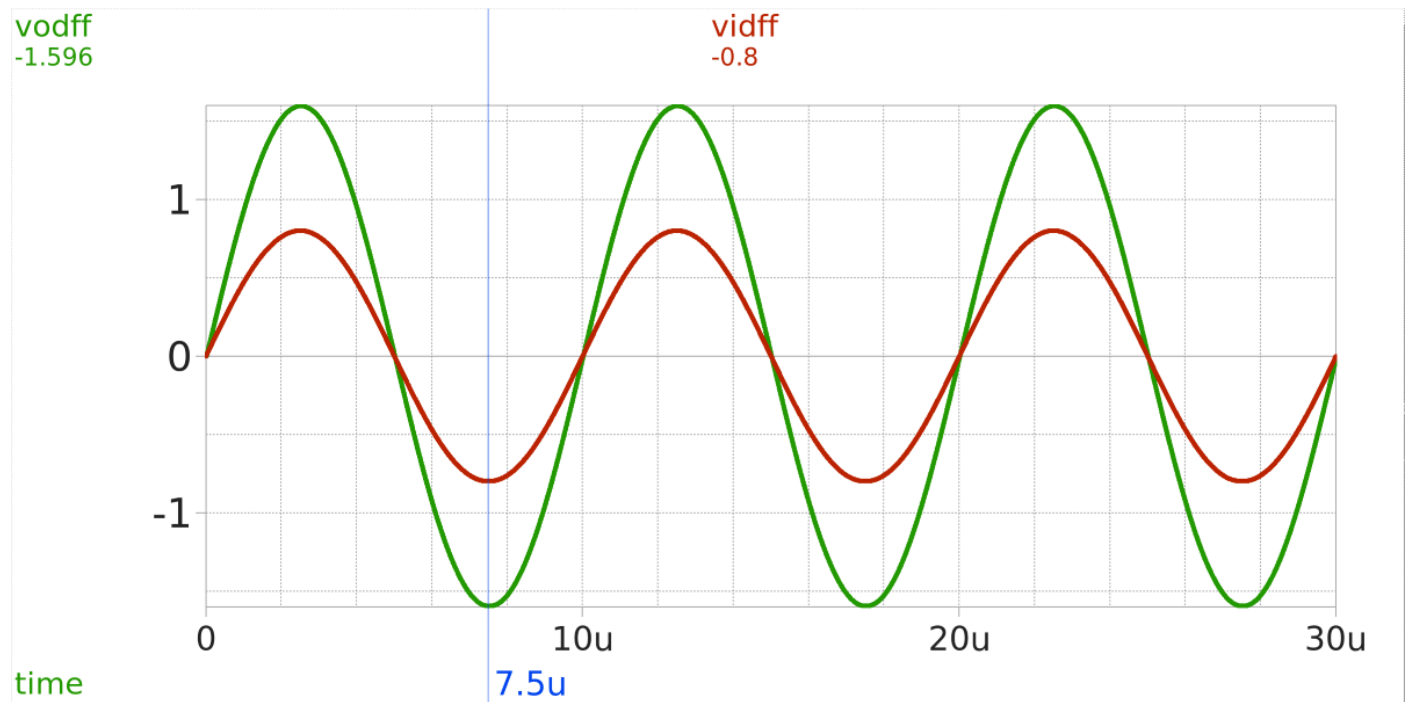


5) Calculate the diff input and output peak-to-peak swings and the closed loop gain.

```
vin_pp = 3.000000e-01  
vout_pp = 5.993790e-01  
gain = 1.997930e+00
```

If we want to know max available swing we should apply a sin with larger amplitude

So let's apply sin with 800mV amplitude



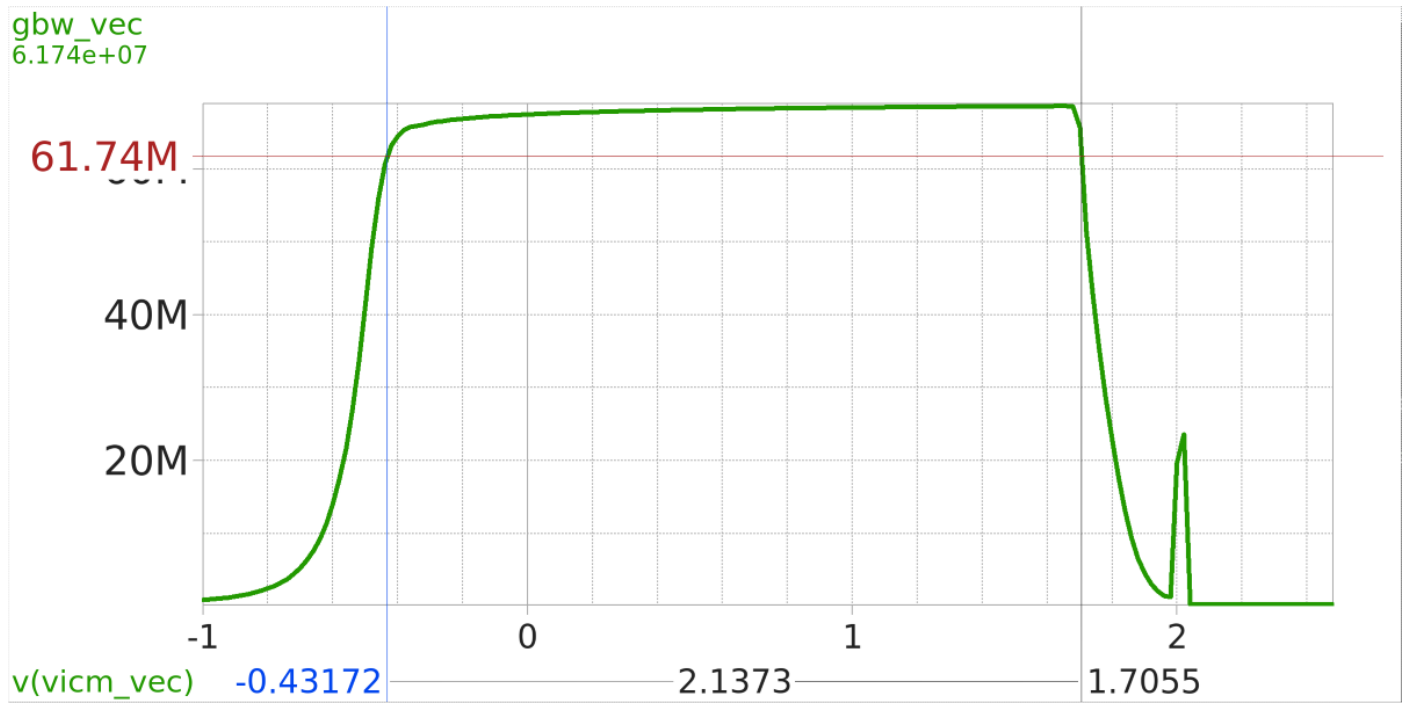
We have output swing = 3.2V p2p

CM Input range using GBW

```
s6
.control
save all
* === DIFFERENTIAL GAIN vs VICM SWEEP ===
let vicm_start = -1
let vicm_stop = 2.5
let vicm_step = 20m
let num_points = (vicm_stop - vicm_start) / vicm_step
* Pre-allocate vectors
let vicm_vec = vector(num_points) *0
let gbw_vec = vector(num_points) *0
* Define vectors type
settype voltage vicm_vec
setscale vicm_vec
settype frequency gbw_vec
let vicm_val = vicm_start
let i = 0
while vicm_val le vicm_stop
* Set CM voltage and enable differential input
alter VICM DC = vicm_val
* AC to get DC gain and BW
acDEC 10 1 10G
meas AC gain MAX vmag(VODFF)
let stop_band=gain*0.707
meas AC bw WHEN vmag(VODFF)=stop_band
* Store results in pre-allocated vectors
let vicm_vec[i] = vicm_val
let gbw_vec[i] = gain * bw
let vicm_val = vicm_val + vicm_step
let i = i + 1
end
* === MEASUREMENTS ===
meas AC max_gbw max f(gbw_vec)
let limit_gbw = 0.9*max_gbw
meas AC min_vincm find v(vicm_vec) when f(gbw_vec)=limit_gbw RISE=1
meas AC max_vincm find v(vicm_vec) when f(gbw_vec)=limit_gbw FALL=1
let CMIR=max_vincm-min_vincm
print min_vincm max_vincm CMIR
write 5t_ota.raw gbw_vec
.endc
```

max_gbw = 6.860859e+07

$$.9 * \text{max GBW} = 61.74 \text{ MHz}$$



Specs summary

	Required	Achieved
Supply voltage	2.5V	
Load	500fF	
Closed loop gain	2	1.998
Phase margin at the required ACL DIFF	$\geq 70^\circ$	89°
Phase margin at the required ACL CM	$\geq 70^\circ$	73.8°
CM input range – low	≤ 0	-0.432V
CM input range – high	$\geq 1V$	1.71V
Differential output swing	1.2Vpk-to-pk	3.2Vpk-to-pk
DC Loop gain	60dB	60dB
CL settling time for 1% error	100ns	80ns